

Plug Puller: the one-sided puller

Quick start, dial guide, measuring guide and measuring form.

Model version 0.13.0

In every picture: black = the tool; teal = your plug; red dotted = the edges this dial moved; the numbers match the key beside the picture.

The dial names are written exactly as the Customizer shows them. Use the bookmarks or the contents page to jump to a part or a dial.

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Quick start

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- `measure_plug_width_cord_end`: Plug width at the cord end
- `measure_plug_thickness_prong_end`: Plug thickness at the prong end
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- `measure_cord_thickness`: Cord thickness
- `measure_wall_plate_style`: Outlet cover plate style
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- `strap_width`: Strap width

Step 4 - Cord Hook

- `hook_hand`: Hook side

Full dial guide

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- `hook_hand`: Hook side

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- `custom_body_top_rounding`: Top edge rounding
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Measuring guide

- `measure_plug_length`: Plug length
- `measure_plug_width_prong_end`: Plug width at the prong end
- `measure_plug_width_cord_end`: Plug width at the cord end
- `measure_plug_thickness_prong_end`: Plug thickness at the prong end
- `measure_plug_thickness_cord_end`: Plug thickness at the cord end
- `measure_cord_thickness`: Cord thickness
- `measure_wall_plate_style`: Outlet cover plate style
- `measure_finger_width`: Finger width
- `measure_hand_width`: Hand width
- `strap_width`: Strap width

Measuring form

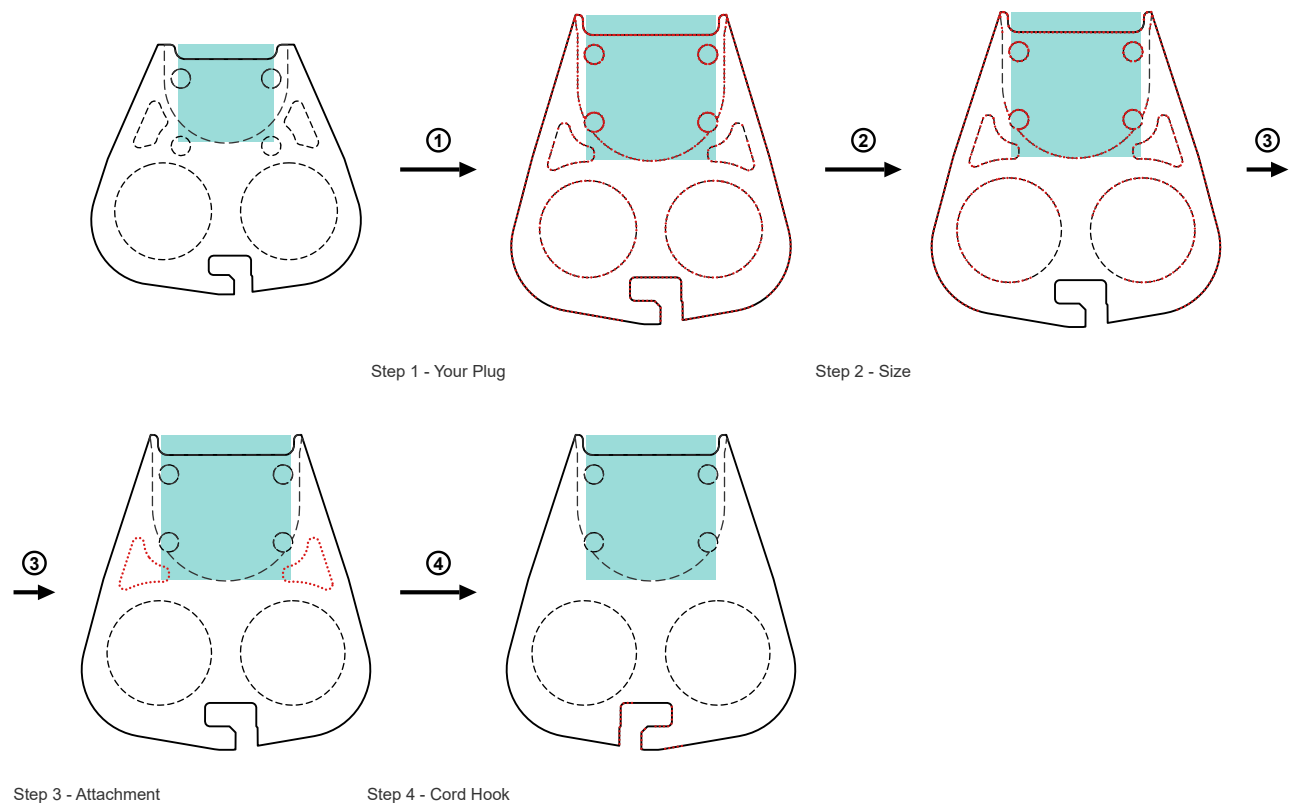
Which tool this is

This packet is for the one-sided puller, src/Plug_Puller_Parametric.scad: the tool for a wall plug up to 24 mm thick, a pocket around the plug's back and sides with two finger holes below it, pulled with one hand. Measure your plug's thickness first. Thicker than 24 mm, or a plug you want held from both sides such as a USB-C tip or a round extension-cord plug, is the two-sided puller's job, and that tool has its own packet.

The Customizer's four steps are your plug, your size, the attachment and the hook's side. The quick start below shows each step's dials, the dial guide every dial of the file, the measuring guide how to take each number, and the measuring form is the sheet you fill in first.

If red text appears beside the part in the preview, read it: it names the measurement to fix, and the part will not fit until it is gone.

In every picture: black = the tool; teal = your plug; red dotted = the edges this dial moved; the numbers match the key beside the picture.



Five stages of the one-sided puller, left to right, the top row first, the plug end at the top; red dots mark the edges each step moved. 1, the defaults: the tool as the file opens, with a 25 mm wide plug in teal. 2, Step 1 - Your Plug: plug length, plug width at the prong end, plug width at the cord end, and 4 more; red on the body edge, the finger holes, the seat, the wall notch and the wing openings. 3, Step 2 - Size: hand size, finger width, hand width; red on the body edge, the finger holes, the pocket, the wall notch, the wing openings and the zip-tie holes. 4, Step 3 - Attachment: attachment; the wing openings removed, drawn in red from the old outline. 5, Step 4 - Cord Hook: hook side; red on the hook.

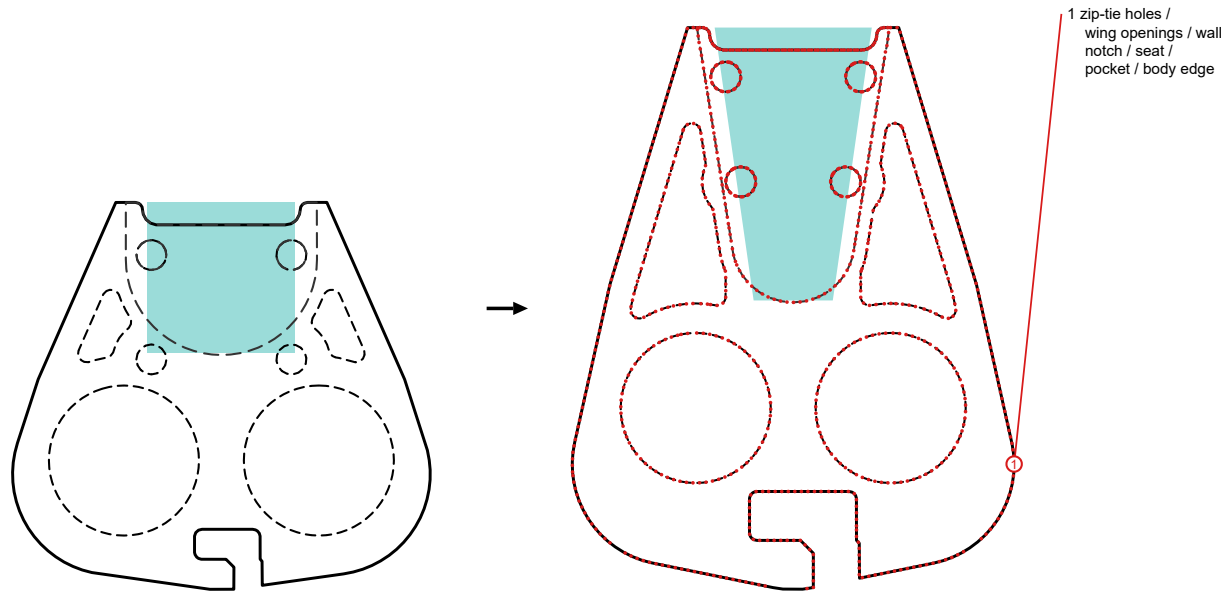
- Step 1: type your plug's numbers
- Step 2: pick your hand size
- Step 3: pick how it attaches
- Step 4: pick the hook's side

Quick start

Step 1 - Your Plug

plug_preset

Plug preset



Measure my plug
→ Standard 3-prong plug - NEMA 5-15

Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 25 mm wide plug in teal. Right: plug preset set to Standard 3-prong plug - NEMA 5-15. Marked in red: 1, the zip-tie holes, the wing openings, the wall notch, the seat, the pocket and the body edge.

Fills in all six plug numbers from a measured reference plug, so the pocket, the wall notch, the cord hook and the zip-tie and wing positions all take that plug's shape at once.

Default Measure my plug · 5 options

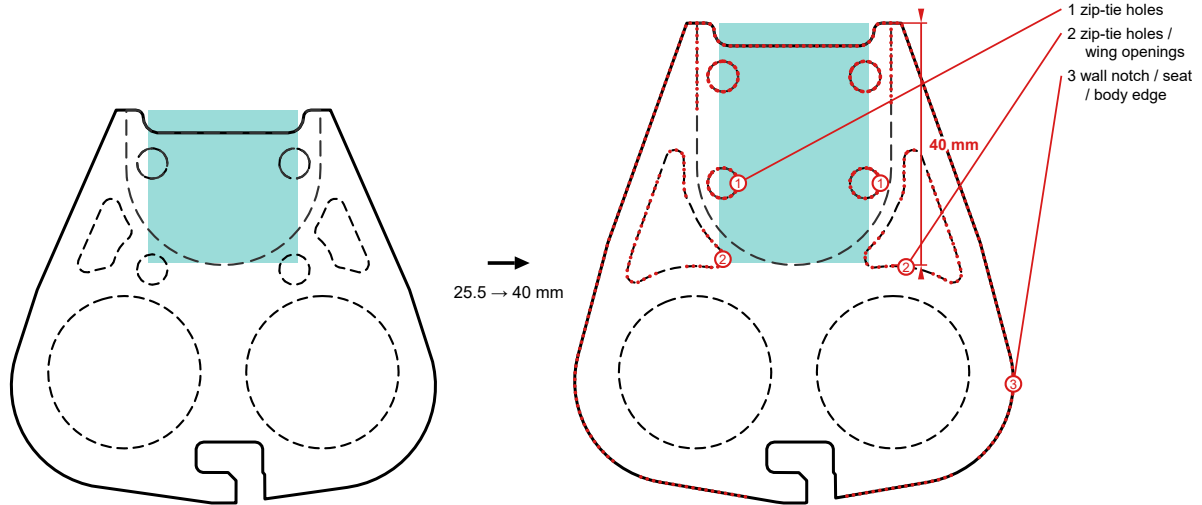
Options: Measure my plug, Flat 2-prong lamp plug - NEMA 1-15, Standard 3-prong plug - NEMA 5-15, Heavy-duty extension cord - NEMA 5-15, Wide 2-prong appliance plug - NEMA 1-15.

Before: Measure my plug. After: Standard 3-prong plug - NEMA 5-15.

Moves: body edge, pocket, seat, wall notch, wing openings, zip-tie holes.

measure_plug_length

Plug length



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 25 mm wide plug in teal. Right: plug length at 40 mm. Marked in red: 1, the zip-tie holes: the four small holes beside the pocket. 2, the zip-tie holes and the wing openings. 3, the wall notch, the seat and the body edge.

Runs the pocket farther toward the finger holes and stretches the whole body to keep the finger holes and zip-tie rows clear of it.

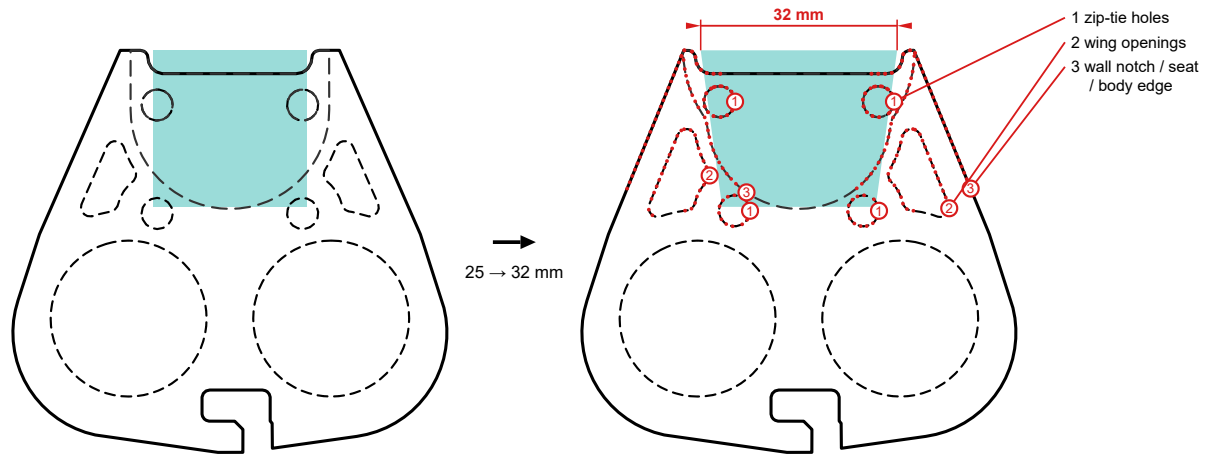
Default 25.5 · Range 12 to 85 · Step size 0.5 · Unit mm

Before: 25.5. After: 40.

Moves: body edge, seat, wall notch, wing openings, zip-tie holes.

measure_plug_width_prong_end

Plug width at the prong end



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 25 mm wide plug in teal. Right: plug width at the prong end at 32 mm. Marked in red: 1, the zip-tie holes: the four small holes beside the pocket. 2, the wing openings: the two openings beside the pocket. 3, the wall notch, the seat and the body edge: the notch in the top edge; the round recess at the plug end; the outer outline.

Widens or narrows the pocket, its seat and the wall notch at the plug end, and moves the zip-tie holes and wing openings with them.

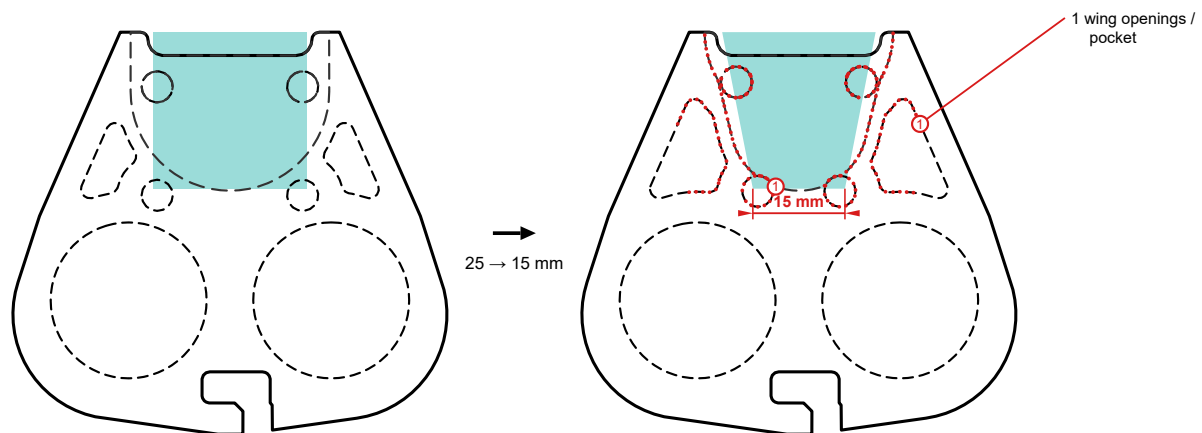
Default 25 · Range 8 to 38 · Step size 0.5 · Unit mm

Before: 25. After: 32.

Moves: body edge, seat, wall notch, wing openings, zip-tie holes.

measure_plug_width_cord_end

Plug width at the cord end



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 25 mm wide plug in teal. Right: plug width at the cord end at 15 mm: tilts the pocket's side walls to match the plug's taper, and the zip-tie holes and wing openings lean in with them. Marked in red: 1, the wing openings and the pocket: the two openings beside the pocket; the plug recess on the centerline.

Tilts the pocket's side walls to match the plug's taper, and the zip-tie holes and wing openings lean in with them.

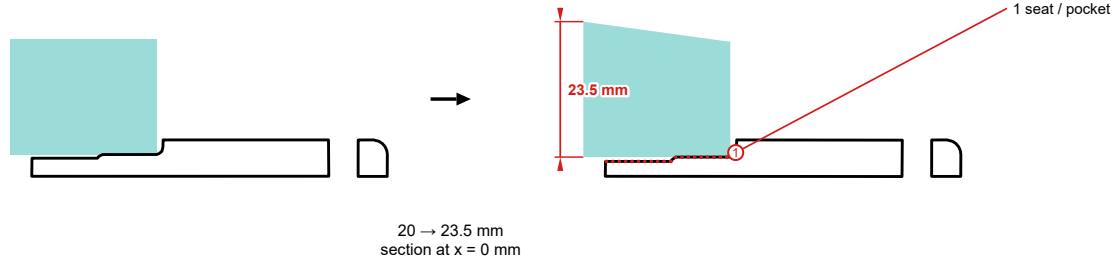
Default 25 · Range 8 to 38 · Step size 0.5 · Unit mm

Before: 25. After: 15.

Moves: pocket, wing openings.

measure_plug_thickness_prong_end

Plug thickness at the prong end



Two vertical slices of the one-sided puller at x = 0 mm, before left and after right, the top face up. Left: the defaults, the plug in teal in the cut. Right: plug thickness at the prong end at 23.5 mm: deepens or shallows the two pocket floors, the round seat and the plug recess, to suit the fatter of the two thickness numbers. Marked in red: 1, the seat and the pocket: the round recess at the plug end; the plug recess on the centerline.

Deepens or shallows the two pocket floors, the round seat and the plug recess, to suit the fatter of the two thickness numbers.

Default 20 · Range 4 to 40 · Step size 0.5 · Unit mm

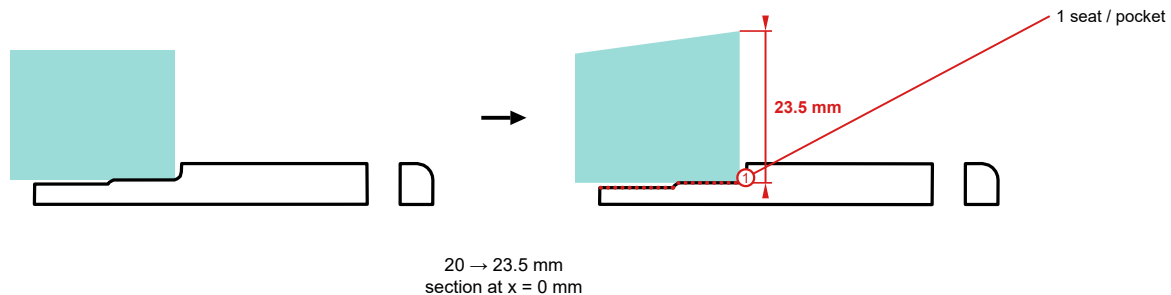
Before: 20. After: 23.5. Section at x = 0 mm.

The fatter of the two thickness numbers sets the floors, so lowering one alone changes nothing; 24 mm or more sends you to the two-sided puller.

Moves: pocket, seat.

measure_plug_thickness_cord_end

Plug thickness at the cord end



Two vertical slices of the one-sided puller at x = 0 mm, before left and after right, the top face up. Left: the defaults, the plug in teal in the cut. Right: plug thickness at the cord end at 23.5 mm: deepens or shallows the two pocket floors, the round seat and the plug recess, when this end is the fatter end of the plug. Marked in red: 1, the seat and the pocket: the round recess at the plug end; the plug recess on the centerline.

Deepens or shallows the two pocket floors, the round seat and the plug recess, when this end is the fatter end of the plug.

Default 20 · Range 4 to 40 · Step size 0.5 · Unit mm

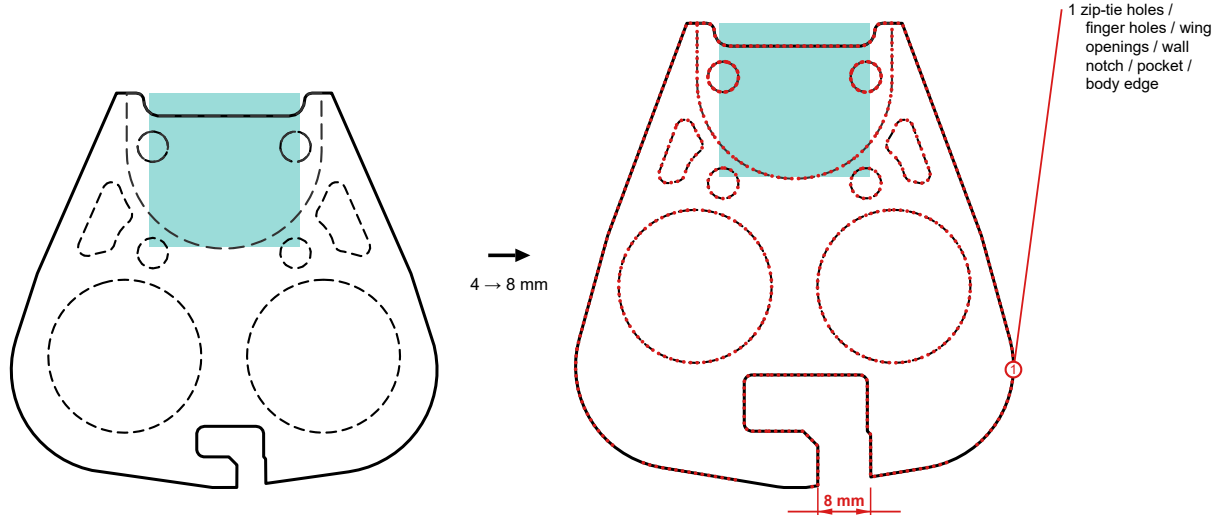
Before: 20. After: 23.5. Section at x = 0 mm.

The fatter of the two thickness numbers sets the floors, so lowering one alone changes nothing; 24 mm or more sends you to the two-sided puller.

Moves: pocket, seat.

measure_cord_thickness

Cord thickness



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 25 mm wide plug in teal. Right: cord thickness at 8 mm. Marked in red: 1, the zip-tie holes, the finger holes, the wing openings, the wall notch, the pocket and the body edge.

Widens the cord hook's slot and crossbar to fit the cord, and moves the finger holes and the whole body up to make room.

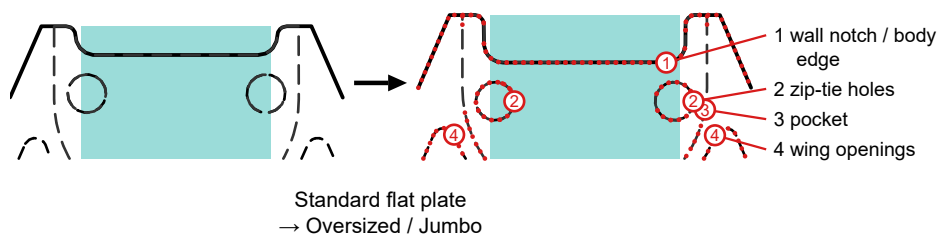
Default 4 · Range 1.5 to 9 · Step size 0.5 · Unit mm

Before: 4. After: 8.

Moves: body edge, finger holes, pocket, wall notch, wing openings, zip-tie holes.

measure_wall_plate_style

Outlet cover plate style



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 25 mm wide plug in teal. Right: outlet cover plate style set to Oversized / Jumbo. Marked in red: 1, the wall notch and the body edge: the notch in the top edge; the outer outline. 2, the zip-tie holes: the four small holes beside the pocket. 3, the pocket: the plug recess on the centerline. 4, the wing openings: the two openings beside the pocket.

Cuts the wall notch at the plug end deeper or shallower to suit the cover plate, and the zip-tie rows shift down with it.

Default Standard flat plate · 4 options

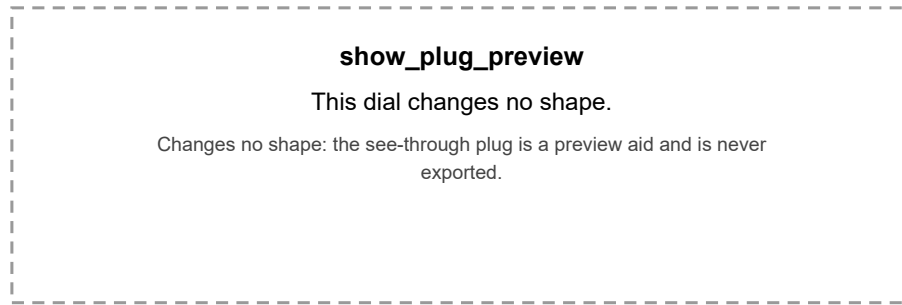
Options: Standard flat plate, Rocker / Decora, Oversized / Jumbo, No plate / flush.

Before: Standard flat plate. After: Oversized / Jumbo.

Moves: body edge, pocket, wall notch, wing openings, zip-tie holes.

show_plug_preview

Show the see-through plug



This dial changes no shape: the see-through plug is a preview aid and is never exported.

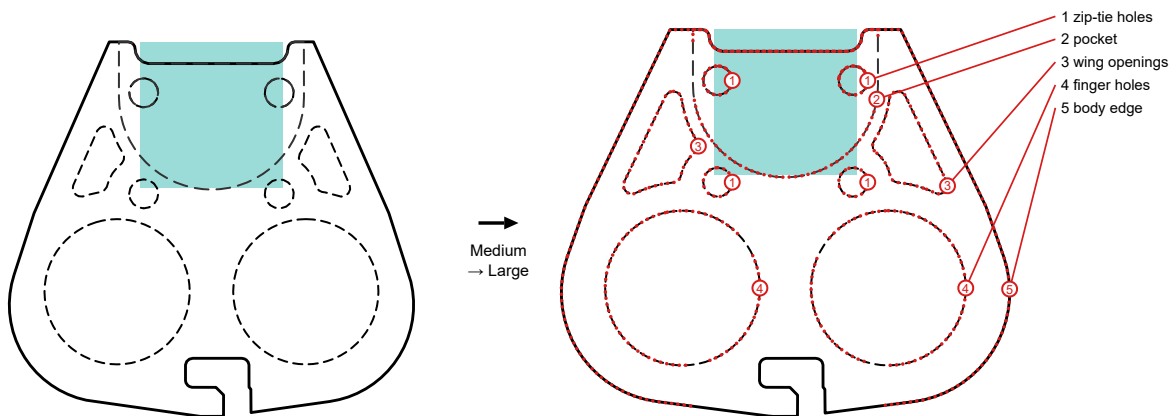
Shows or hides the see-through plug in the preview and changes nothing in the printed tool.

Default true · true / false

Step 2 - Size

size

Hand size



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 25 mm wide plug in teal. Right: hand size set to Large. Marked in red: 1, the zip-tie holes: the four small holes beside the pocket. 2, the pocket: the plug recess on the centerline. 3, the wing openings: the two openings beside the pocket. 4, the finger holes: the two large holes in the lower half. 5, the body edge: the outer outline.

Scales the whole body, both finger holes, the wing openings and the zip-tie grid to the chosen hand size.

Default Medium · 5 options

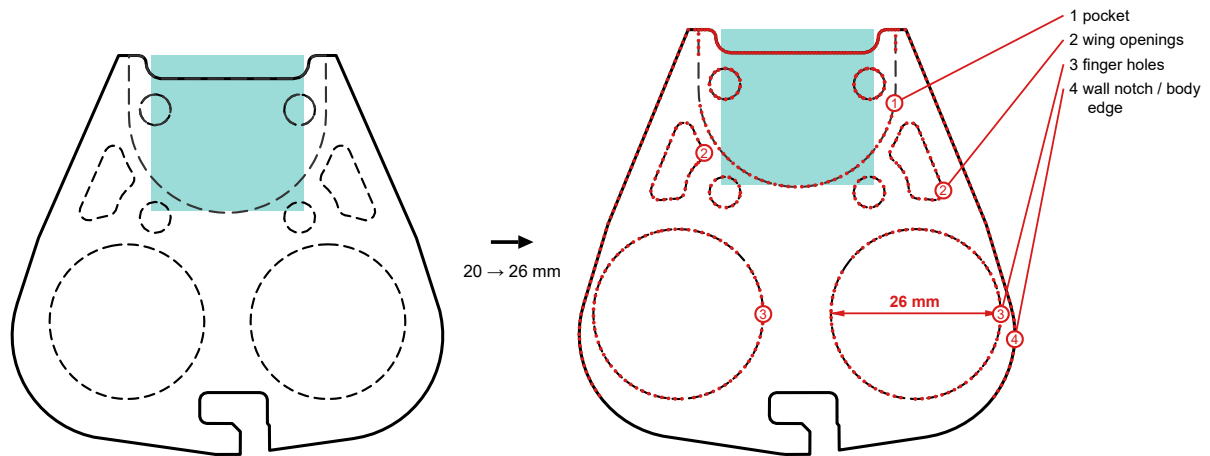
Options: Small, Medium, Large, Measure my hand, Custom.

Before: Medium. After: Large.

Moves: body edge, finger holes, pocket, wing openings, zip-tie holes.

measure_finger_width

Finger width



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Measure my hand, a 25 mm wide plug in teal. Right: finger width at 26 mm. Marked in red: 1, the pocket: the plug recess on the centerline. 2, the wing openings: the two openings beside the pocket. 3, the finger holes: the two large holes in the lower half. 4, the wall notch and the body edge.

Widens both finger holes, spaces them farther apart and pushes them and the body up to keep the walls printable.

Default 20 · Range 14 to 32 · Step size 0.5 · Unit mm

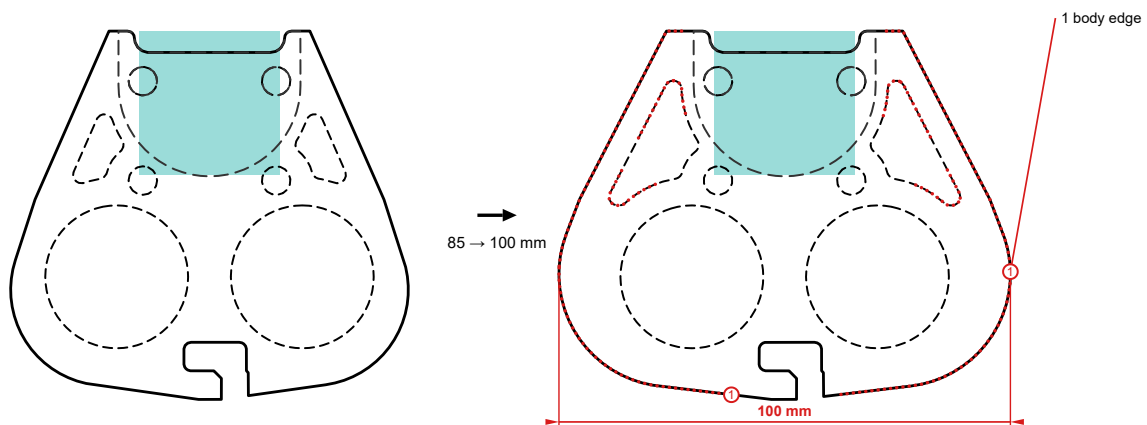
Before: 20. After: 26. Context: size = Measure my hand.

Only acts when size is Measure my hand.

Moves: body edge, finger holes, pocket, wall notch, wing openings.

measure_hand_width

Hand width



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Measure my hand, a 25 mm wide plug in teal. Right: hand width at 100 mm: widens the whole body outline and thickens the slab in step with the hand. Marked in red: 1, the body edge: the outer outline. Everything else follows the outline.

Widens the whole body outline and thickens the slab in step with the hand.

Default 85 · Range 60 to 110 · Step size 1 · Unit mm

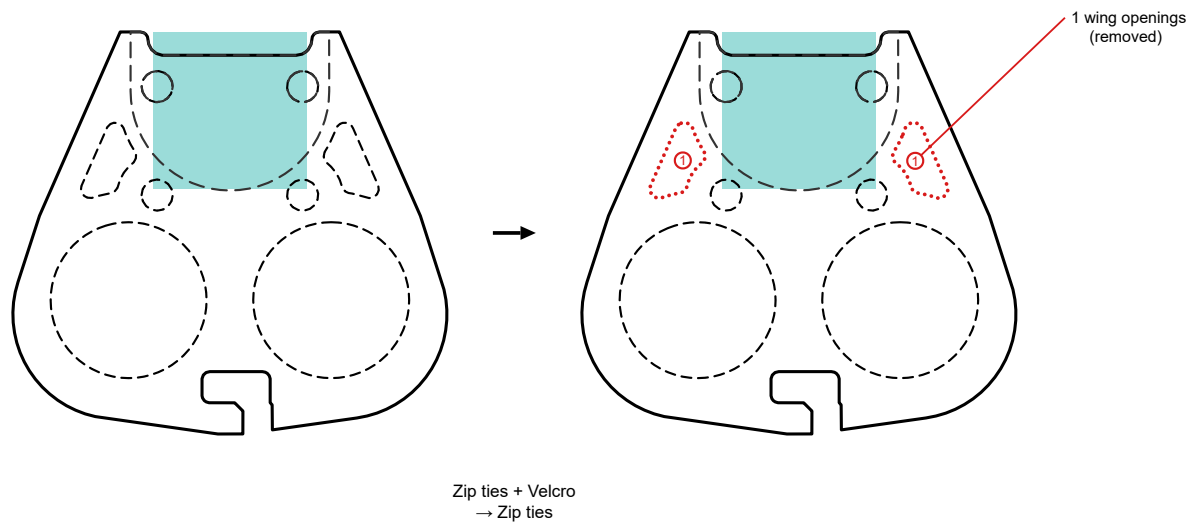
Before: 85. After: 100. Context: size = Measure my hand.

Only acts when size is Measure my hand.

Moves: body edge.

Step 3 - Attachment

attachment Attachment



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 25 mm wide plug in teal. Right: attachment set to Zip ties: adds or removes the zip-tie holes and the wing or slot openings for the strap. Marked in red: 1, the wing openings: the two openings beside the pocket, removed. The body edge, the pocket, the seat and the wall notch stay where they were.

Adds or removes the zip-tie holes and the wing or slot openings for the strap.

Default Zip ties + Velcro · 4 options

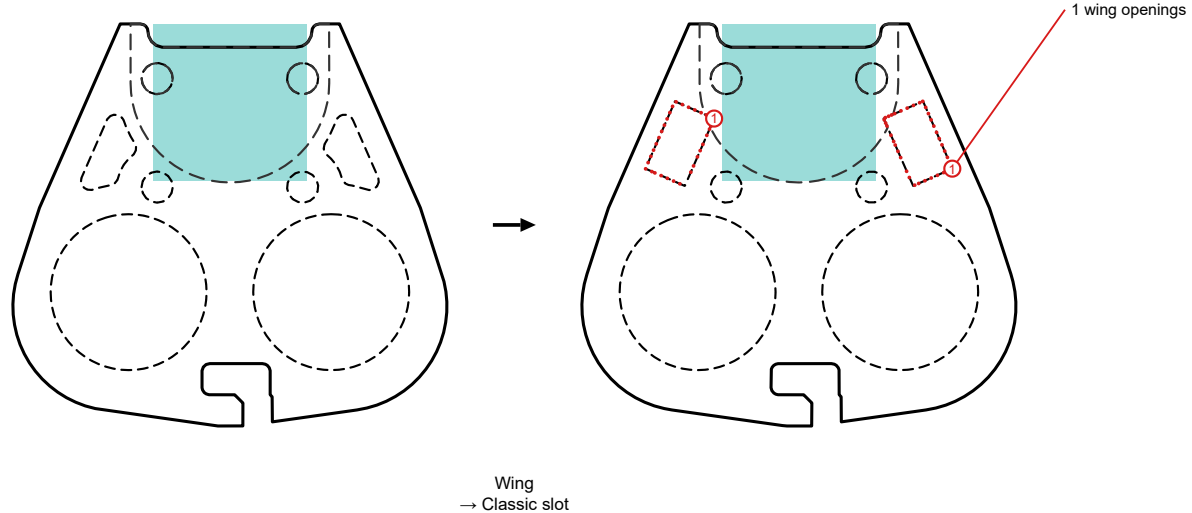
Options: Zip ties, Velcro strap, Zip ties + Velcro, None.

Before: Zip ties + Velcro. After: Zip ties.

Moves: wing openings.

velcro_style

Strap opening style



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 25 mm wide plug in teal. Right: strap opening style set to Classic slot: swaps the curved wing openings for a pair of plain rectangular slots that lean along the body's sides. Marked in red: 1, the wing openings: the two openings beside the pocket. The body edge, the pocket, the seat and the wall notch stay where they were.

Swaps the curved wing openings for a pair of plain rectangular slots that lean along the body's sides.

Default Wing · 2 options

Options: Wing, Classic slot.

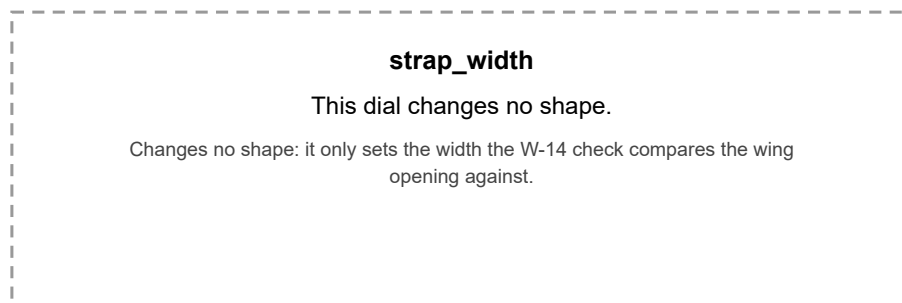
Before: Wing. After: Classic slot.

Moves: wing openings.

Can trip: WING OPENING SMALLER THAN STRAP WIDTH.

strap_width

Strap width



This dial changes no shape: it only sets the width the W-14 check compares the wing opening against.

Sets the strap width the wing opening is checked against and changes no shape.

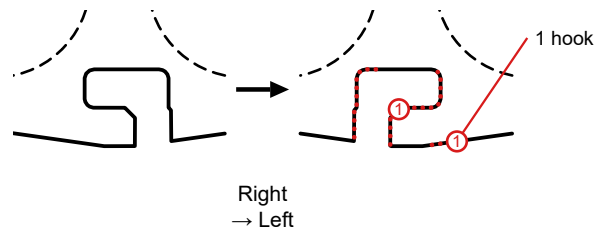
Default 15 · Range 10 to 25 · Step size 1 · Unit mm

Can trip: WING OPENING SMALLER THAN STRAP WIDTH; WING WEB COLLAPSED - NO ROOM FOR STRAP.

Step 4 - Cord Hook

hook_hand

Hook side



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 25 mm wide plug in teal. Right: hook side set to Left: mirrors the cord hook at the cord end so its catch faces the other side, and nothing else moves. Marked in red: 1, the hook: the cord hook slot in the bottom edge. The body edge, the pocket, the seat and the wall notch stay where they were.

Mirrors the cord hook at the cord end so its catch faces the other side, and nothing else moves.

Default Right · 2 options

Options: Right, Left.

Before: Right. After: Left.

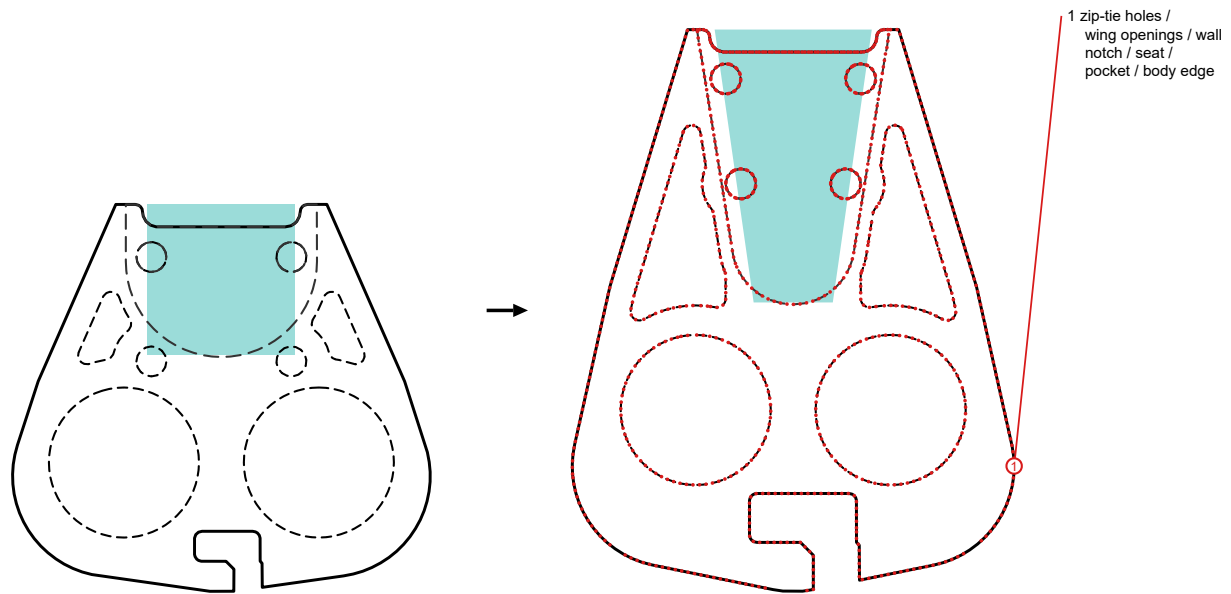
Moves: hook.

Full dial guide

Step 1 - Your Plug

plug_preset

Plug preset



Measure my plug
→ Standard 3-prong plug - NEMA 5-15

Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 25 mm wide plug in teal. Right: plug preset set to Standard 3-prong plug - NEMA 5-15. Marked in red: 1, the zip-tie holes, the wing openings, the wall notch, the seat, the pocket and the body edge.

Fills in all six plug numbers from a measured reference plug, so the pocket, the wall notch, the cord hook and the zip-tie and wing positions all take that plug's shape at once.

Default Measure my plug · 5 options

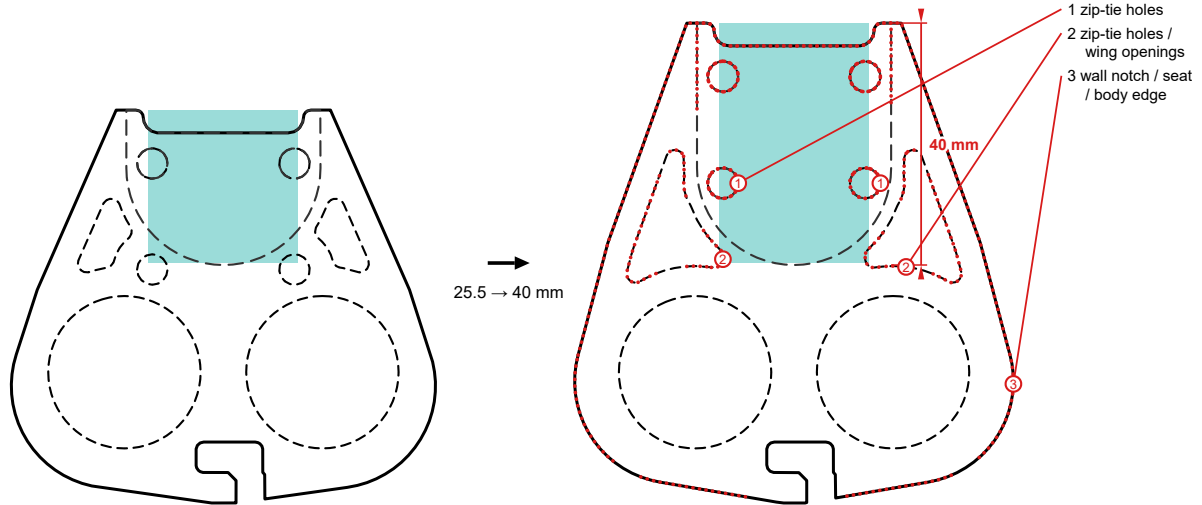
Options: Measure my plug, Flat 2-prong lamp plug - NEMA 1-15, Standard 3-prong plug - NEMA 5-15, Heavy-duty extension cord - NEMA 5-15, Wide 2-prong appliance plug - NEMA 1-15.

Before: Measure my plug. After: Standard 3-prong plug - NEMA 5-15.

Moves: body edge, pocket, seat, wall notch, wing openings, zip-tie holes.

measure_plug_length

Plug length



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 25 mm wide plug in teal. Right: plug length at 40 mm. Marked in red: 1, the zip-tie holes: the four small holes beside the pocket. 2, the zip-tie holes and the wing openings. 3, the wall notch, the seat and the body edge.

Runs the pocket farther toward the finger holes and stretches the whole body to keep the finger holes and zip-tie rows clear of it.

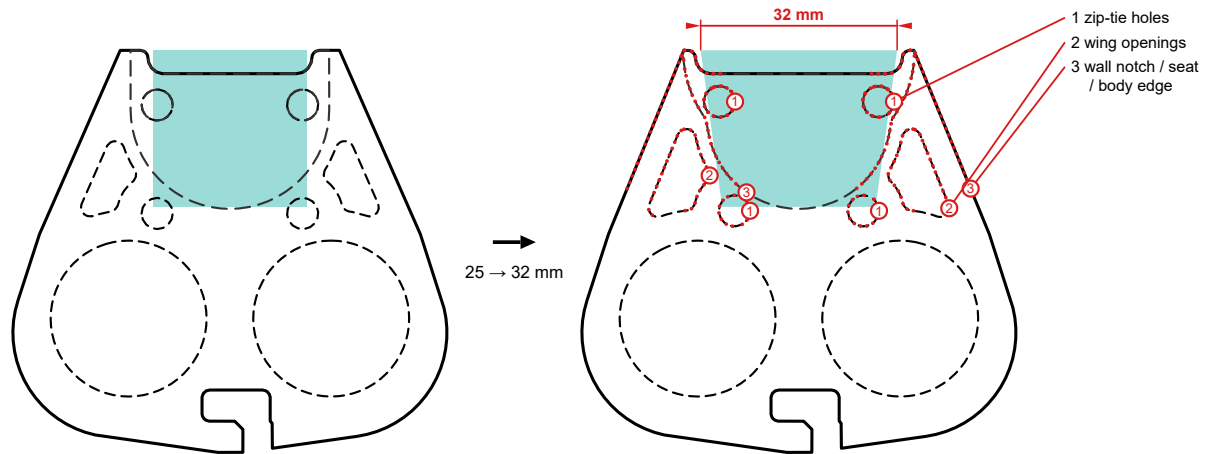
Default 25.5 · Range 12 to 85 · Step size 0.5 · Unit mm

Before: 25.5. After: 40.

Moves: body edge, seat, wall notch, wing openings, zip-tie holes.

measure_plug_width_prong_end

Plug width at the prong end



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 25 mm wide plug in teal. Right: plug width at the prong end at 32 mm. Marked in red: 1, the zip-tie holes: the four small holes beside the pocket. 2, the wing openings: the two openings beside the pocket. 3, the wall notch, the seat and the body edge: the notch in the top edge; the round recess at the plug end; the outer outline.

Widens or narrows the pocket, its seat and the wall notch at the plug end, and moves the zip-tie holes and wing openings with them.

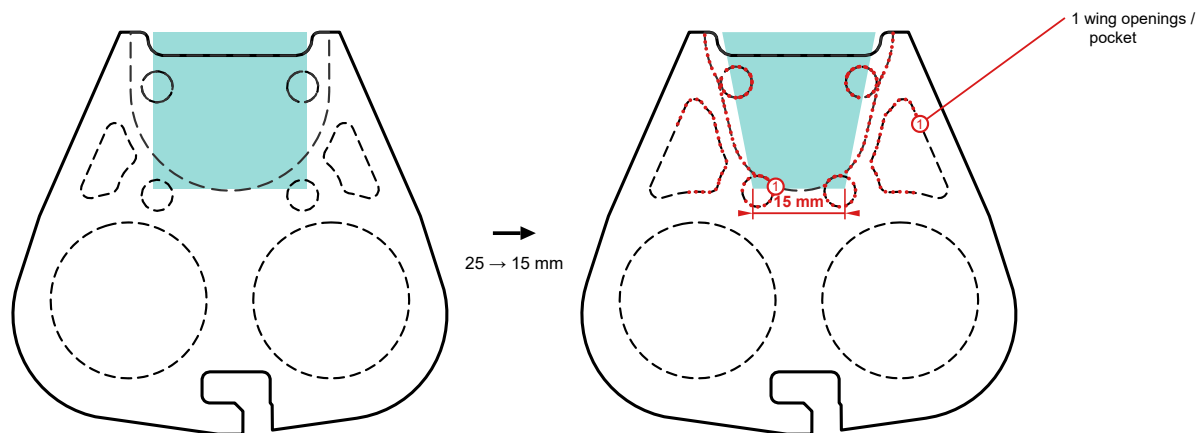
Default 25 · Range 8 to 38 · Step size 0.5 · Unit mm

Before: 25. After: 32.

Moves: body edge, seat, wall notch, wing openings, zip-tie holes.

measure_plug_width_cord_end

Plug width at the cord end



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 25 mm wide plug in teal. Right: plug width at the cord end at 15 mm: tilts the pocket's side walls to match the plug's taper, and the zip-tie holes and wing openings lean in with them. Marked in red: 1, the wing openings and the pocket: the two openings beside the pocket; the plug recess on the centerline.

Tilts the pocket's side walls to match the plug's taper, and the zip-tie holes and wing openings lean in with them.

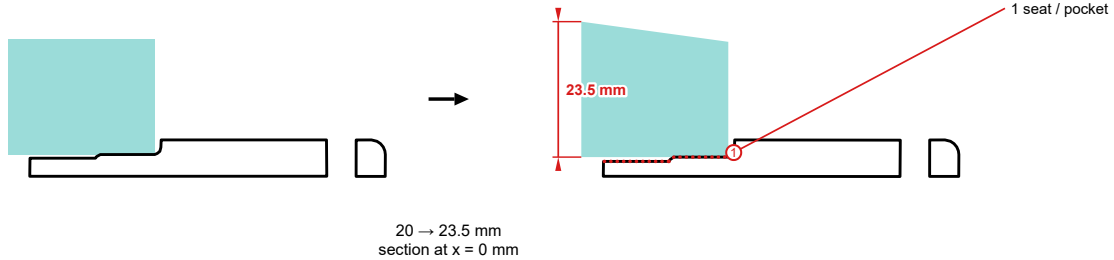
Default 25 · Range 8 to 38 · Step size 0.5 · Unit mm

Before: 25. After: 15.

Moves: pocket, wing openings.

measure_plug_thickness_prong_end

Plug thickness at the prong end



Two vertical slices of the one-sided puller at x = 0 mm, before left and after right, the top face up. Left: the defaults, the plug in teal in the cut. Right: plug thickness at the prong end at 23.5 mm: deepens or shallows the two pocket floors, the round seat and the plug recess, to suit the fatter of the two thickness numbers. Marked in red: 1, the seat and the pocket: the round recess at the plug end; the plug recess on the centerline.

Deepens or shallows the two pocket floors, the round seat and the plug recess, to suit the fatter of the two thickness numbers.

Default 20 · Range 4 to 40 · Step size 0.5 · Unit mm

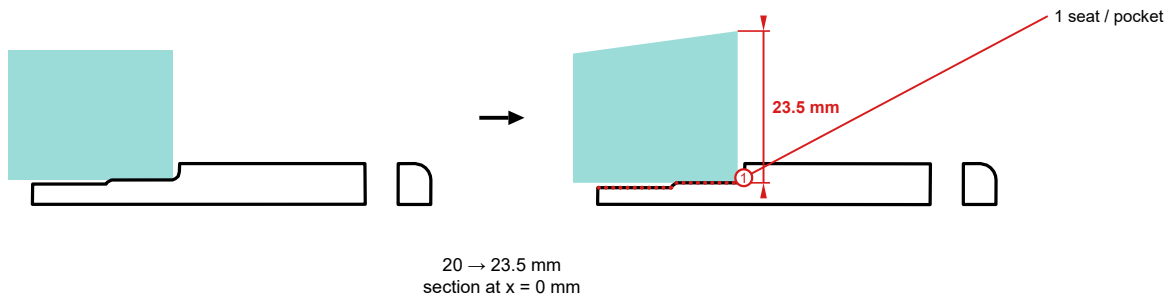
Before: 20. After: 23.5. Section at x = 0 mm.

The fatter of the two thickness numbers sets the floors, so lowering one alone changes nothing; 24 mm or more sends you to the two-sided puller.

Moves: pocket, seat.

measure_plug_thickness_cord_end

Plug thickness at the cord end



Two vertical slices of the one-sided puller at x = 0 mm, before left and after right, the top face up. Left: the defaults, the plug in teal in the cut. Right: plug thickness at the cord end at 23.5 mm: deepens or shallows the two pocket floors, the round seat and the plug recess, when this end is the fatter end of the plug. Marked in red: 1, the seat and the pocket: the round recess at the plug end; the plug recess on the centerline.

Deepens or shallows the two pocket floors, the round seat and the plug recess, when this end is the fatter end of the plug.

Default 20 · Range 4 to 40 · Step size 0.5 · Unit mm

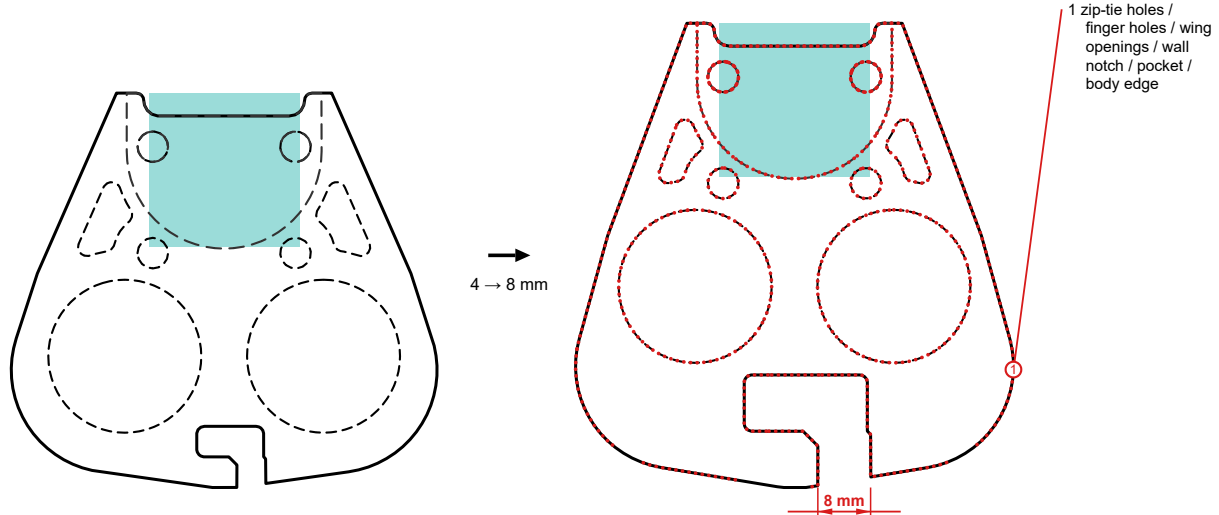
Before: 20. After: 23.5. Section at x = 0 mm.

The fatter of the two thickness numbers sets the floors, so lowering one alone changes nothing; 24 mm or more sends you to the two-sided puller.

Moves: pocket, seat.

measure_cord_thickness

Cord thickness



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 25 mm wide plug in teal. Right: cord thickness at 8 mm. Marked in red: 1, the zip-tie holes, the finger holes, the wing openings, the wall notch, the pocket and the body edge.

Widens the cord hook's slot and crossbar to fit the cord, and moves the finger holes and the whole body up to make room.

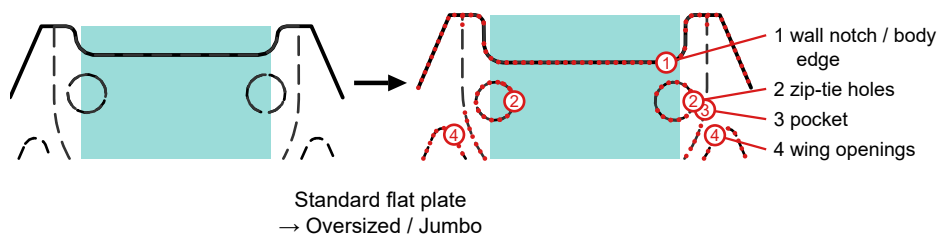
Default 4 · Range 1.5 to 9 · Step size 0.5 · Unit mm

Before: 4. After: 8.

Moves: body edge, finger holes, pocket, wall notch, wing openings, zip-tie holes.

measure_wall_plate_style

Outlet cover plate style



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 25 mm wide plug in teal. Right: outlet cover plate style set to Oversized / Jumbo. Marked in red: 1, the wall notch and the body edge: the notch in the top edge; the outer outline. 2, the zip-tie holes: the four small holes beside the pocket. 3, the pocket: the plug recess on the centerline. 4, the wing openings: the two openings beside the pocket.

Cuts the wall notch at the plug end deeper or shallower to suit the cover plate, and the zip-tie rows shift down with it.

Default Standard flat plate · 4 options

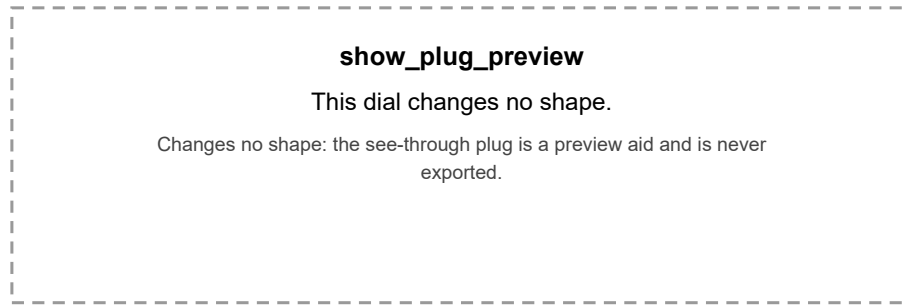
Options: Standard flat plate, Rocker / Decora, Oversized / Jumbo, No plate / flush.

Before: Standard flat plate. After: Oversized / Jumbo.

Moves: body edge, pocket, wall notch, wing openings, zip-tie holes.

show_plug_preview

Show the see-through plug



This dial changes no shape: the see-through plug is a preview aid and is never exported.

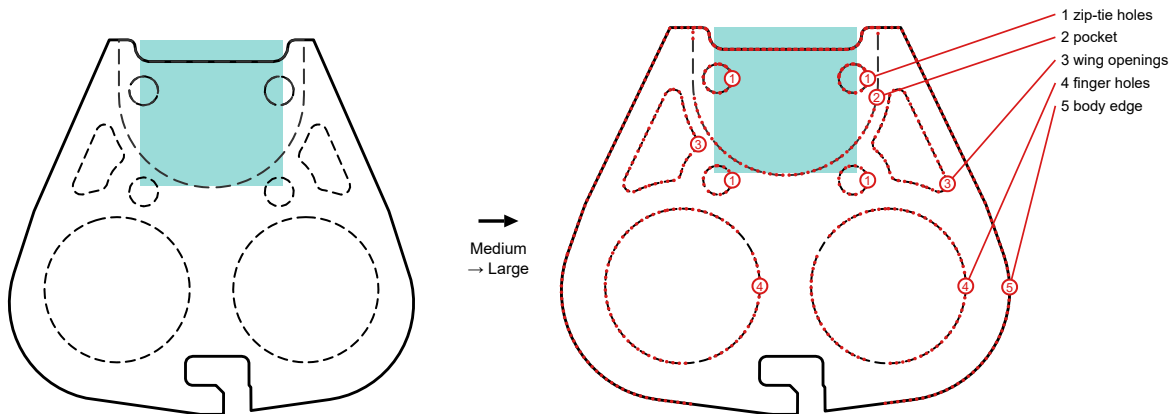
Shows or hides the see-through plug in the preview and changes nothing in the printed tool.

Default true · true / false

Step 2 - Size

size

Hand size



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 25 mm wide plug in teal. Right: hand size set to Large. Marked in red: 1, the zip-tie holes: the four small holes beside the pocket. 2, the pocket: the plug recess on the centerline. 3, the wing openings: the two openings beside the pocket. 4, the finger holes: the two large holes in the lower half. 5, the body edge: the outer outline.

Scales the whole body, both finger holes, the wing openings and the zip-tie grid to the chosen hand size.

Default Medium · 5 options

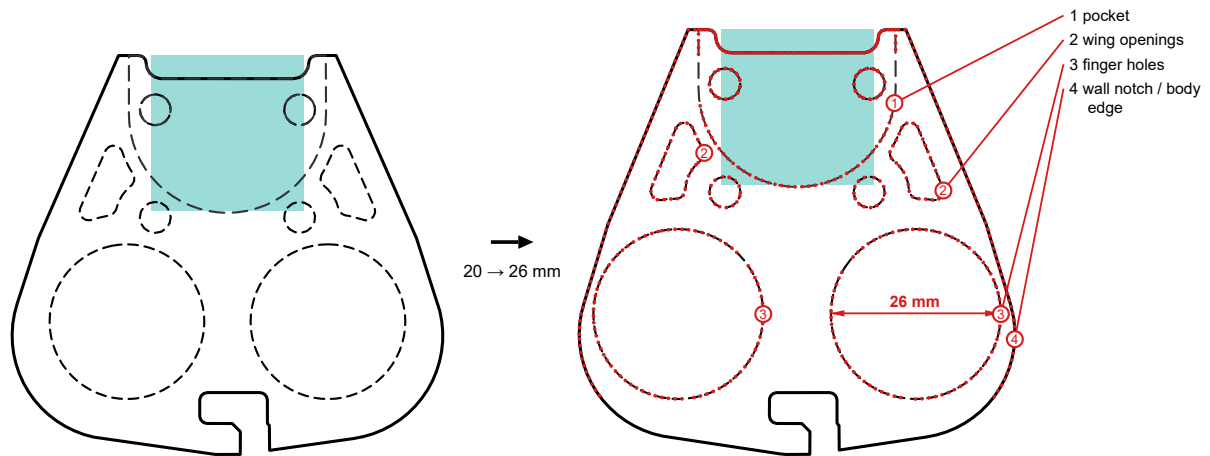
Options: Small, Medium, Large, Measure my hand, Custom.

Before: Medium. After: Large.

Moves: body edge, finger holes, pocket, wing openings, zip-tie holes.

measure_finger_width

Finger width



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Measure my hand, a 25 mm wide plug in teal. Right: finger width at 26 mm. Marked in red: 1, the pocket: the plug recess on the centerline. 2, the wing openings: the two openings beside the pocket. 3, the finger holes: the two large holes in the lower half. 4, the wall notch and the body edge.

Widens both finger holes, spaces them farther apart and pushes them and the body up to keep the walls printable.

Default 20 · Range 14 to 32 · Step size 0.5 · Unit mm

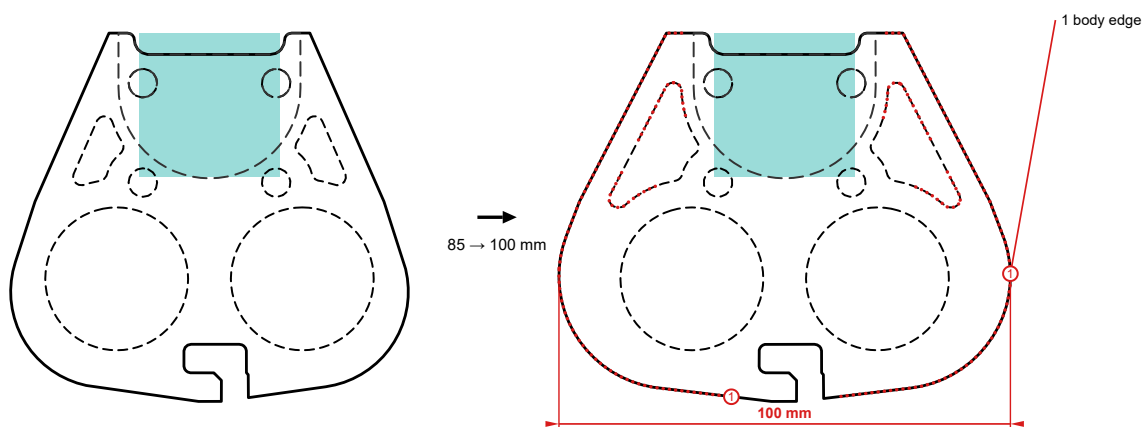
Before: 20. After: 26. Context: size = Measure my hand.

Only acts when size is Measure my hand.

Moves: body edge, finger holes, pocket, wall notch, wing openings.

measure_hand_width

Hand width



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Measure my hand, a 25 mm wide plug in teal. Right: hand width at 100 mm: widens the whole body outline and thickens the slab in step with the hand. Marked in red: 1, the body edge: the outer outline. Everything else follows the outline.

Widens the whole body outline and thickens the slab in step with the hand.

Default 85 · Range 60 to 110 · Step size 1 · Unit mm

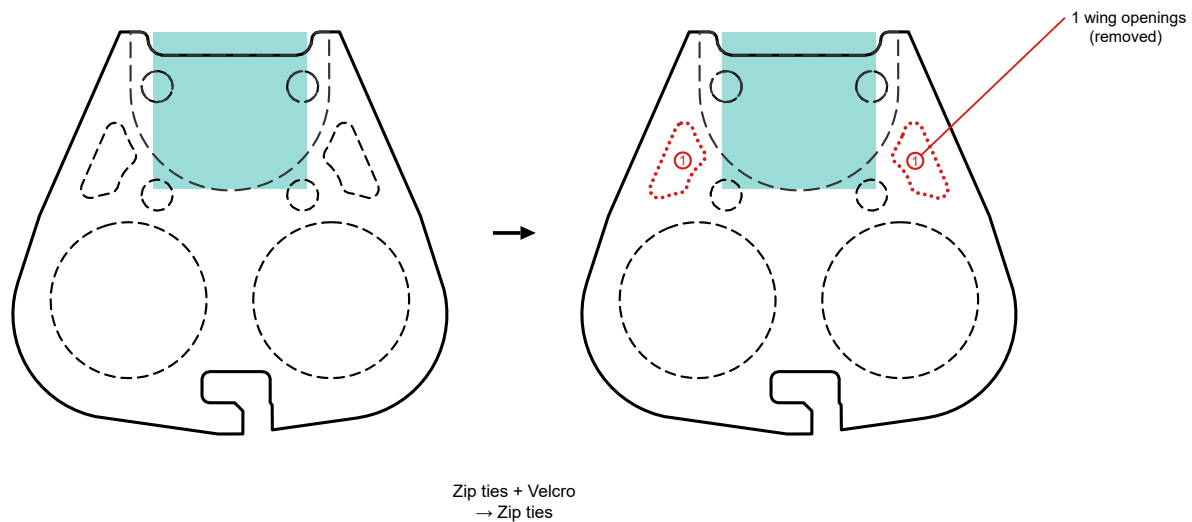
Before: 85. After: 100. Context: size = Measure my hand.

Only acts when size is Measure my hand.

Moves: body edge.

Step 3 - Attachment

attachment Attachment



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 25 mm wide plug in teal. Right: attachment set to Zip ties: adds or removes the zip-tie holes and the wing or slot openings for the strap. Marked in red: 1, the wing openings: the two openings beside the pocket, removed. The body edge, the pocket, the seat and the wall notch stay where they were.

Adds or removes the zip-tie holes and the wing or slot openings for the strap.

Default Zip ties + Velcro · 4 options

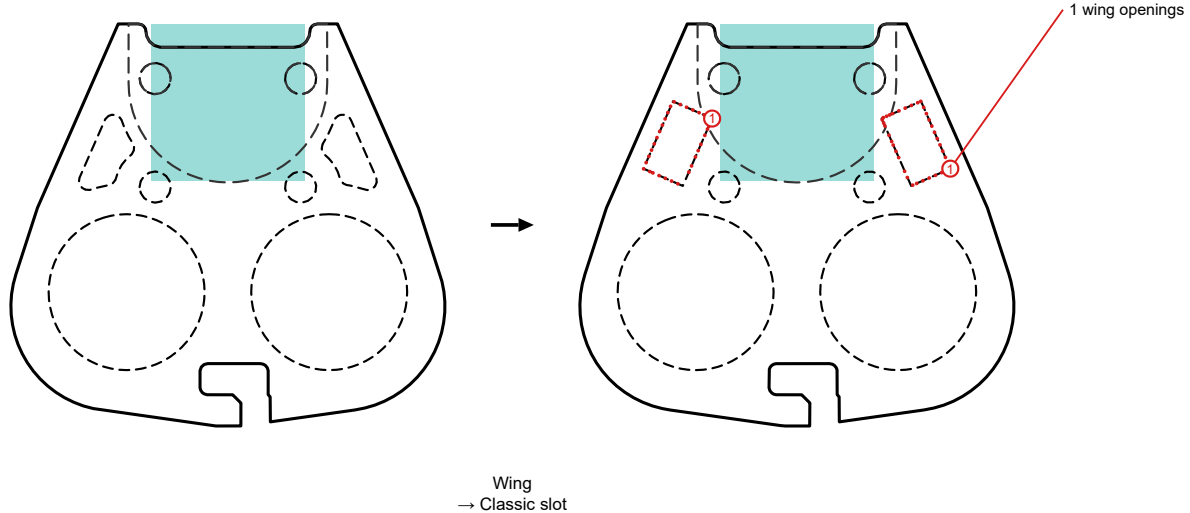
Options: Zip ties, Velcro strap, Zip ties + Velcro, None.

Before: Zip ties + Velcro. After: Zip ties.

Moves: wing openings.

velcro_style

Strap opening style



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 25 mm wide plug in teal. Right: strap opening style set to Classic slot: swaps the curved wing openings for a pair of plain rectangular slots that lean along the body's sides. Marked in red: 1, the wing openings: the two openings beside the pocket. The body edge, the pocket, the seat and the wall notch stay where they were.

Swaps the curved wing openings for a pair of plain rectangular slots that lean along the body's sides.

Default Wing · 2 options

Options: Wing, Classic slot.

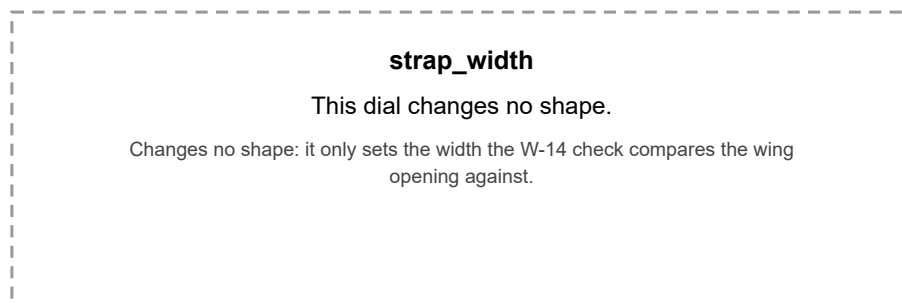
Before: Wing. After: Classic slot.

Moves: wing openings.

Can trip: WING OPENING SMALLER THAN STRAP WIDTH.

strap_width

Strap width



This dial changes no shape: it only sets the width the W-14 check compares the wing opening against.

Sets the strap width the wing opening is checked against and changes no shape.

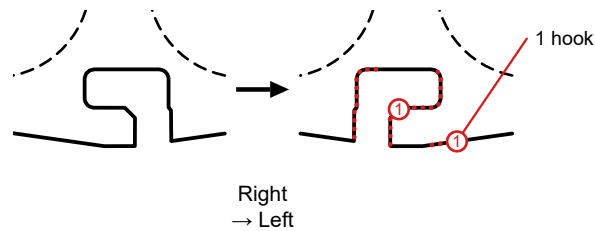
Default 15 · Range 10 to 25 · Step size 1 · Unit mm

Can trip: WING OPENING SMALLER THAN STRAP WIDTH; WING WEB COLLAPSED - NO ROOM FOR STRAP.

Step 4 - Cord Hook

hook_hand

Hook side



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 25 mm wide plug in teal. Right: hook side set to Left: mirrors the cord hook at the cord end so its catch faces the other side, and nothing else moves. Marked in red: 1, the hook: the cord hook slot in the bottom edge. The body edge, the pocket, the seat and the wall notch stay where they were.

Mirrors the cord hook at the cord end so its catch faces the other side, and nothing else moves.

Default Right · 2 options

Options: Right, Left.

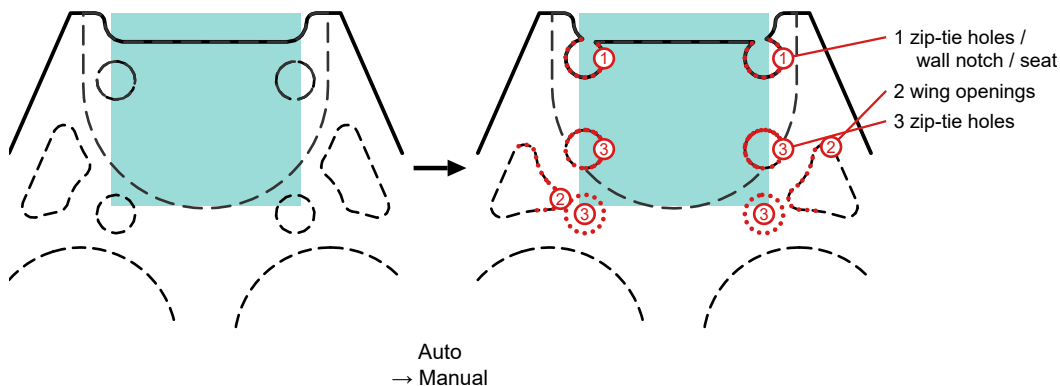
Before: Right. After: Left.

Moves: hook.

Advanced - Zip Tie Placement

zip_placement

Zip-tie hole placement



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 25 mm wide plug in teal. Right: zip-tie hole placement set to Manual. Marked in red: 1, the zip-tie holes, the wall notch and the seat: the four small holes beside the pocket; the notch in the top edge; the round recess at the plug end. 2, the wing openings: the two openings beside the pocket. 3, the zip-tie holes: the four small holes beside the pocket.

Switches the zip-tie pairs from the automatic rows to the three position dials, which slide each pair along the plug's side.

Default Auto · 2 options

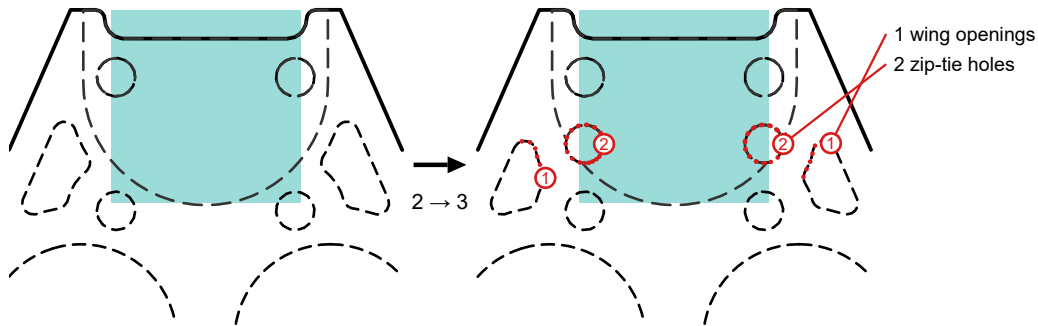
Options: Auto, Manual.

Before: Auto. After: Manual.

Moves: seat, wall notch, wing openings, zip-tie holes.

zip_row_count

Number of zip-tie rows



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 25 mm wide plug in teal. Right: number of zip-tie rows at 3: adds or removes a pair of zip-tie holes, and the rows squeeze together when the body has no room for another. Marked in red: 1, the wing openings: the two openings beside the pocket. 2, the zip-tie holes: the four small holes beside the pocket.

Adds or removes a pair of zip-tie holes, and the rows squeeze together when the body has no room for another.

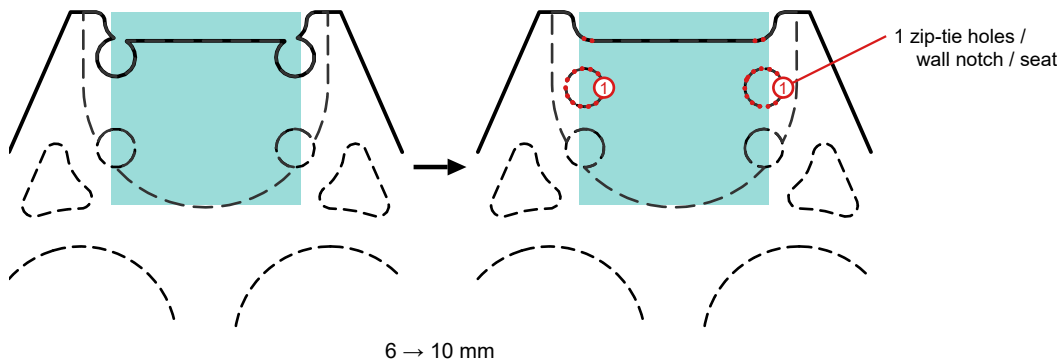
Default 2 · Range 1 to 3 · Step size 1 · Unit —

Before: 2. After: 3.

Moves: wing openings, zip-tie holes.

zip_pos_1

Zip-tie pair 1 position



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with zip-tie hole placement set to Manual, a 25 mm wide plug in teal. Right: zip-tie pair 1 position at 10 mm. Marked in red: 1, the zip-tie holes, the wall notch and the seat: the four small holes beside the pocket; the notch in the top edge; the round recess at the plug end.

Slides the first pair of zip-tie holes along the plug's side, measured from the plug end.

Default 6 · Range 0 to 55 · Step size 0.5 · Unit mm

Before: 6. After: 10. Context: zip_placement = Manual.

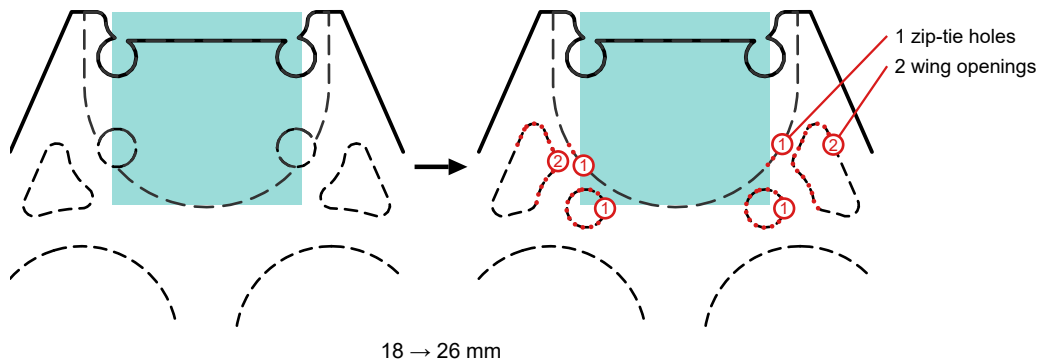
Only acts when zip_placement is Manual.

Moves: seat, wall notch, zip-tie holes.

Can trip: ZIP TIE HOLES HIT FINGER HOLES; ZIP TIE ROWS OVERLAP EACH OTHER; ZIP TIE HOLES HIT VELCRO SLOTS.

zip_pos_2

Zip-tie pair 2 position



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with zip-tie hole placement set to Manual, a 25 mm wide plug in teal. Right: zip-tie pair 2 position at 26 mm: slides the second pair of zip-tie holes along the plug's side, measured from the plug end. Marked in red: 1, the zip-tie holes: the four small holes beside the pocket. 2, the wing openings: the two openings beside the pocket.

Slides the second pair of zip-tie holes along the plug's side, measured from the plug end.

Default 18 · Range 0 to 55 · Step size 0.5 · Unit mm

Before: 18. After: 26. Context: zip_placement = Manual.

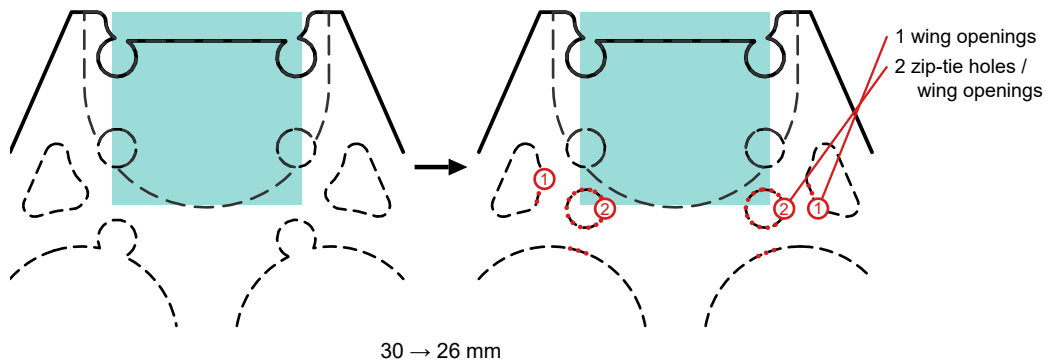
Only acts when zip_placement is Manual.

Moves: wing openings, zip-tie holes.

Can trip: ZIP TIE HOLES HIT FINGER HOLES; ZIP TIE ROWS OVERLAP EACH OTHER; ZIP TIE HOLES HIT VELCRO SLOTS.

zip_pos_3

Zip-tie pair 3 position



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with zip-tie hole placement set to Manual and number of zip-tie rows set to 3, a 25 mm wide plug in teal. Right: zip-tie pair 3 position at 26 mm. Marked in red: 1, the wing openings: the two openings beside the pocket. 2, the zip-tie holes and the wing openings: the four small holes beside the pocket; the two openings beside the pocket.

Slides the third pair of zip-tie holes along the plug's side, measured from the plug end.

Default 30 · Range 0 to 55 · Step size 0.5 · Unit mm

Before: 30. After: 26. Context: zip_placement = Manual, zip_row_count = 3.

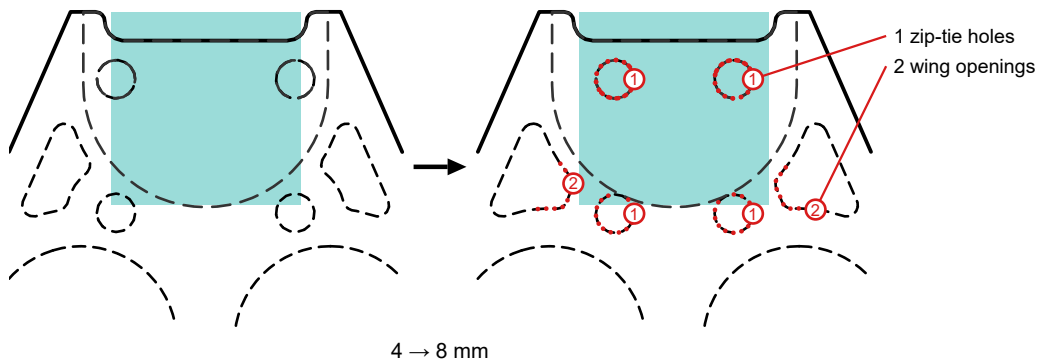
Only acts when zip_placement is Manual and zip_row_count is 3.

Moves: wing openings, zip-tie holes.

Can trip: ZIP TIE HOLES HIT FINGER HOLES; ZIP TIE ROWS OVERLAP EACH OTHER; ZIP TIE HOLES HIT VELCRO SLOTS.

zip_edge_offset

Zip-tie hole distance from the pocket wall



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 25 mm wide plug in teal. Right: zip-tie hole distance from the pocket wall at 8 mm: moves every zip-tie hole farther from or closer to the pocket wall, and the wing openings reshape around them. Marked in red: 1, the zip-tie holes: the four small holes beside the pocket. 2, the wing openings: the two openings beside the pocket.

Moves every zip-tie hole farther from or closer to the pocket wall, and the wing openings reshape around them.

Default 4 · Range 2.5 to 12 · Step size 0.25 · Unit mm

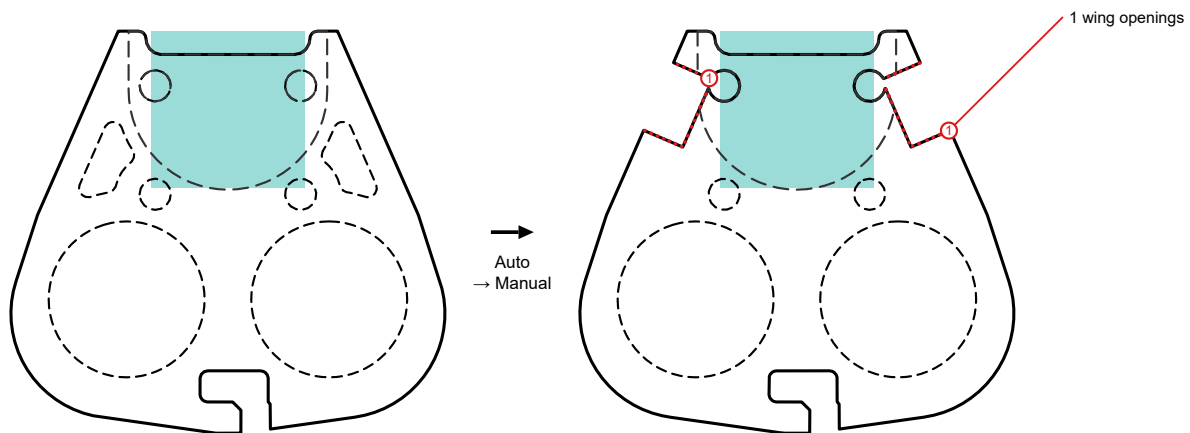
Before: 4. After: 8.

Moves: wing openings, zip-tie holes.

Advanced - Velcro Placement

velcro_placement

Strap slot placement



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 25 mm wide plug in teal. Right: strap slot placement set to Manual: switches the strap openings to a pair of plain slots that the position dial slides along the plug's side, whatever the opening style. Marked in red: 1, the wing openings: the two openings beside the pocket. The body edge, the pocket, the seat and the wall notch stay where they were.

Switches the strap openings to a pair of plain slots that the position dial slides along the plug's side, whatever the opening style.

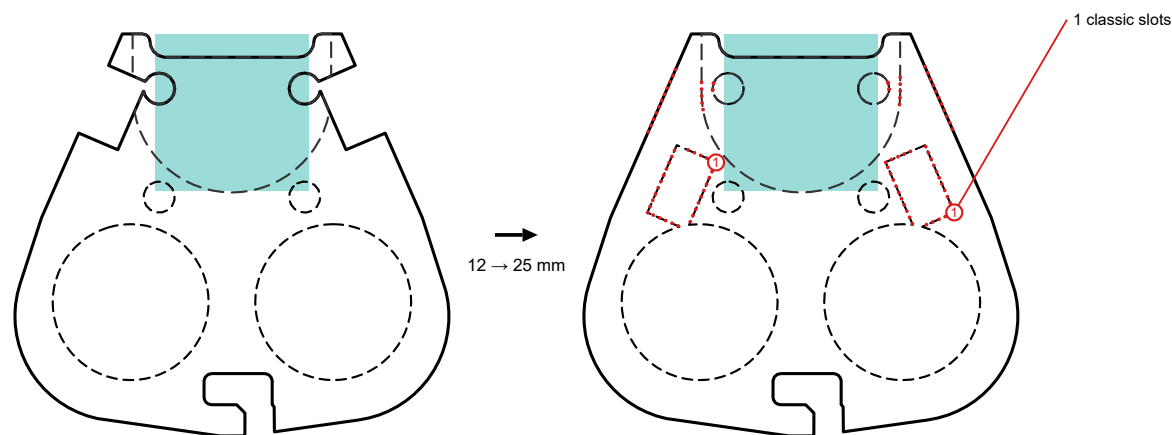
Default Auto · 2 options

Options: Auto, Manual.

Before: Auto. After: Manual.

Moves: wing openings.

velcro_pos
Strap slot position



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with strap opening style set to Classic slot and strap slot placement set to Manual, a 25 mm wide plug in teal. Right: strap slot position at 25 mm: slides the pair of strap slots along the plug's side, measured from the plug end. Marked in red: 1, the classic slots: the two slots beside the pocket.

Slides the pair of strap slots along the plug's side, measured from the plug end.

Default 12 · Range 0 to 55 · Step size 0.5 · Unit mm

Before: 12. After: 25. Context: velcro_style = Classic slot, velcro_placement = Manual.

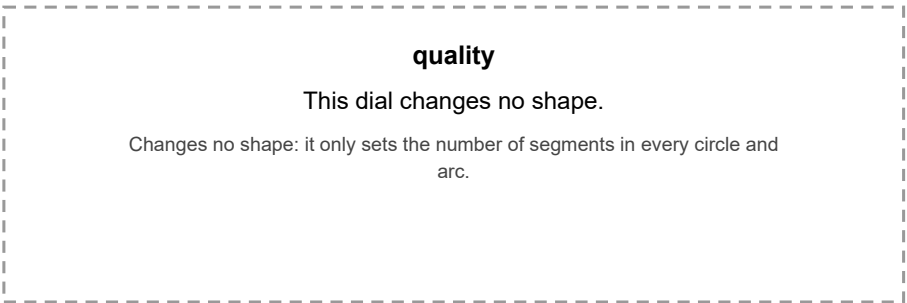
Only acts when velcro_placement is Manual.

Moves: classic slots.

Can trip: ZIP TIE HOLES HIT VELCRO SLOTS.

Advanced - Render Quality

quality
Render quality



This dial changes no shape: it only sets the number of segments in every circle and arc.

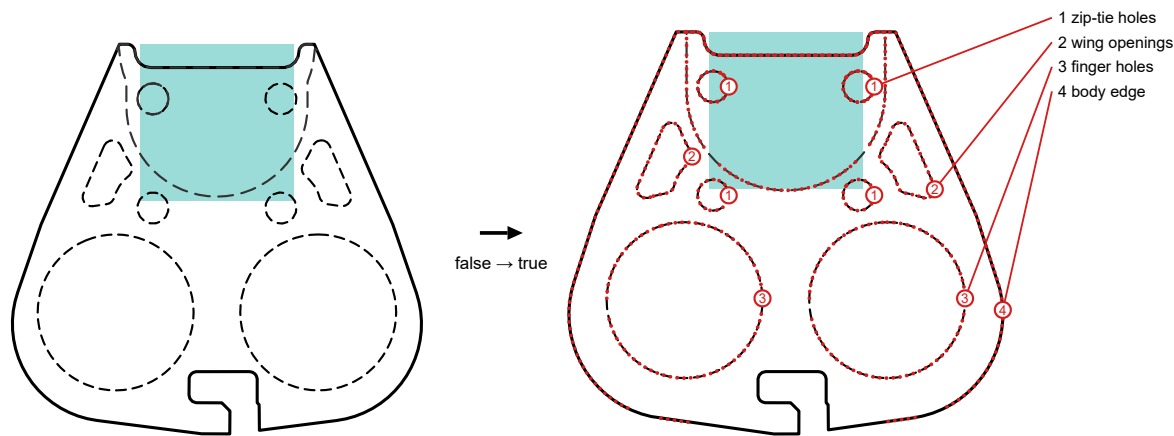
Sets how many flat segments draw each curve and changes no dimension.

Default 64 · Range 24 to 128 · Step size 8 · Unit segments

Custom Mode

reset_custom_to_medium

Reset Custom to the Medium reference



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom, a 25 mm wide plug in teal. Right: reset Custom to the Medium reference set to true. Marked in red: 1, the zip-tie holes: the four small holes beside the pocket. 2, the wing openings: the two openings beside the pocket. 3, the finger holes: the two large holes in the lower half. 4, the body edge: the outer outline.

Swaps every Custom slider for the Medium reference values for one render, so the whole tool snaps back to the Medium shape.

Default false · true / false

Before: false. After: true. Context: size = Custom.

Only acts when size is Custom.

Moves: body edge, finger holes, wing openings, zip-tie holes.

custom_enable_auto_fit

Auto-fit

custom_enable_auto_fit

This dial changes no shape.

Changes no shape at the Custom defaults: it only clamps sliders that leave their safe range, and every clamp is reported in the console.

This dial changes no shape. Changes no shape at the Custom defaults: it only clamps sliders that leave their safe range, and every clamp is reported in the console.

Lets the tool clamp any Custom slider that would push a feature past the body or into a neighbor, and changes nothing while every slider is inside its safe range.

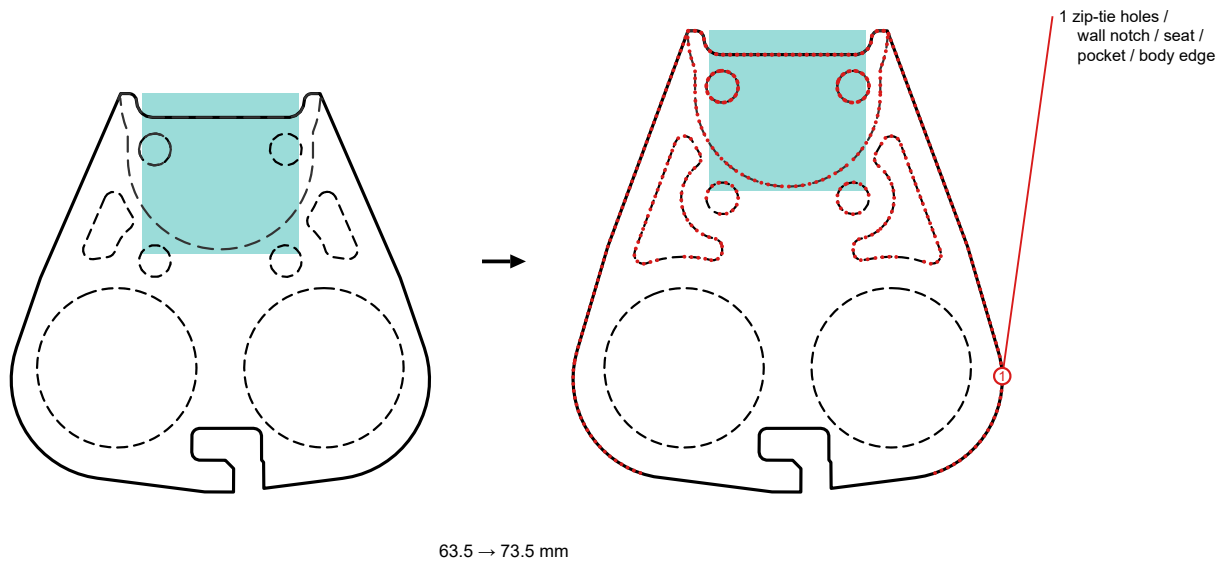
Default true · true / false

Can trip: ZIP TIE HOLES HIT FINGER HOLES; FINGER HOLES OUTSIDE BODY; POCKET SEAT WIDER THAN TOP EDGE; POCKET WIDER THAN BODY; WALL NOTCH WIDER THAN TOP EDGE.

Body Shape (Custom size only)

custom_puller_length

Body length



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom, a 25 mm wide plug in teal. Right: body length at 73.5 mm. Marked in red: 1, the zip-tie holes, the wall notch, the seat, the pocket and the body edge: the four small holes beside the pocket; the notch in the top edge; the round recess at the plug end; the plug recess on the centerline; the outer outline.

Stretches the whole body from the cord end to the plug end.

Default 63.5 · Range 50 to 120 · Step size 0.5 · Unit mm

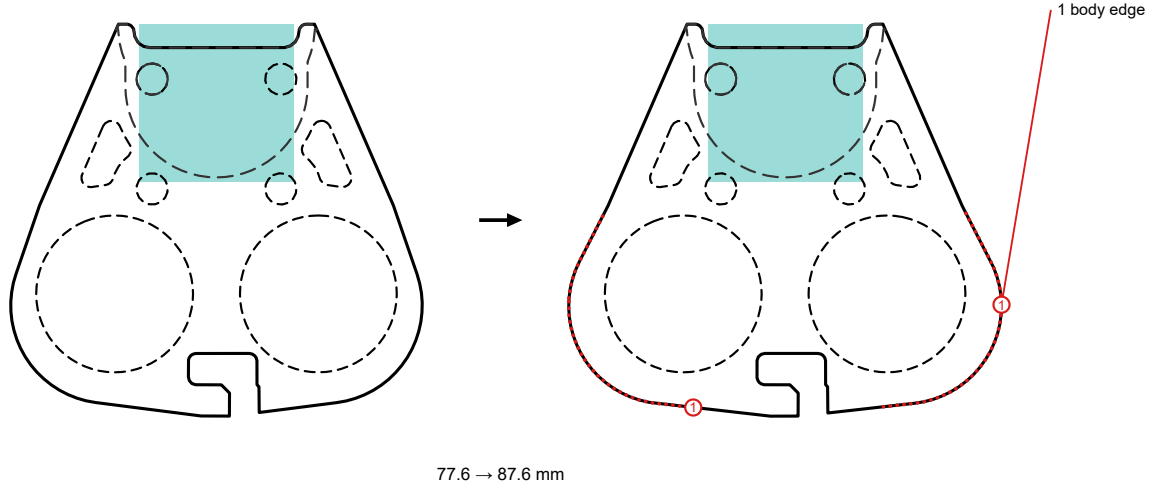
Before: 63.5. After: 73.5. Context: size = Custom.

Only acts when size is Custom.

Moves: body edge, pocket, seat, wall notch, zip-tie holes.

custom_puller_bottom_width

Body width at the widest point



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom, a 25 mm wide plug in teal. Right: body width at the widest point at 87.6 mm: widens the body at its widest point, just above the cord end, before the side rounding is applied. Marked in red: 1, the body edge: the outer outline. Everything else follows the outline.

Widens the body at its widest point, just above the cord end, before the side rounding is applied.

Default 77.6 · Range 50 to 120 · Step size 0.05 · Unit mm

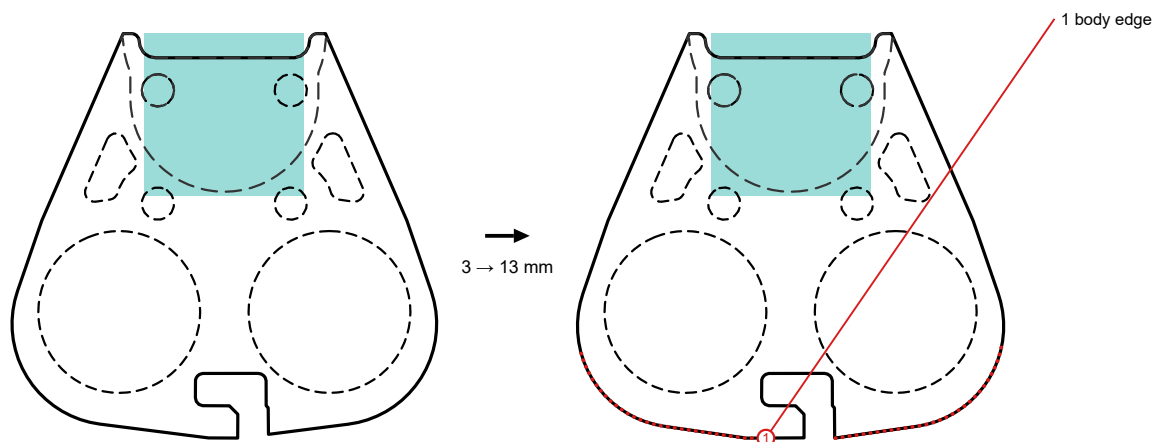
Before: 77.6. After: 87.6. Context: size = Custom.

Only acts when size is Custom.

Moves: body edge.

custom_puller_bottom_corners

Cord-end corner width



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom, a 25 mm wide plug in teal. Right: cord-end corner width at 13 mm: widens the flat at the cord end where the two lower corners of the body outline sit. Marked in red: 1, the body edge: the outer outline. Everything else follows the outline.

Widens the flat at the cord end where the two lower corners of the body outline sit.

Default 3 · Range 3 to 80 · Step size 0.25 · Unit mm

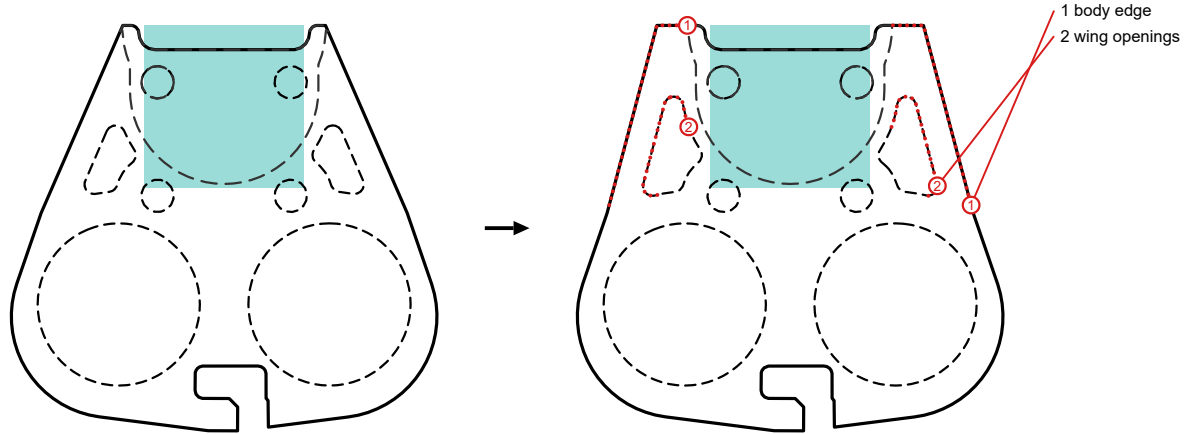
Before: 3. After: 13. Context: size = Custom.

Only acts when size is Custom.

Moves: body edge.

custom_puller_top_width

Body width at the plug end



31.75 → 41.75 mm

Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom, a 25 mm wide plug in teal. Right: body width at the plug end at 41.75 mm: widens the plug end of the body, where the wall notch and the pocket seat sit. Marked in red: 1, the body edge: the outer outline. 2, the wing openings: the two openings beside the pocket. Everything else follows the outline.

Widens the plug end of the body, where the wall notch and the pocket seat sit.

Default 31.75 · Range 25 to 80 · Step size 0.25 · Unit mm

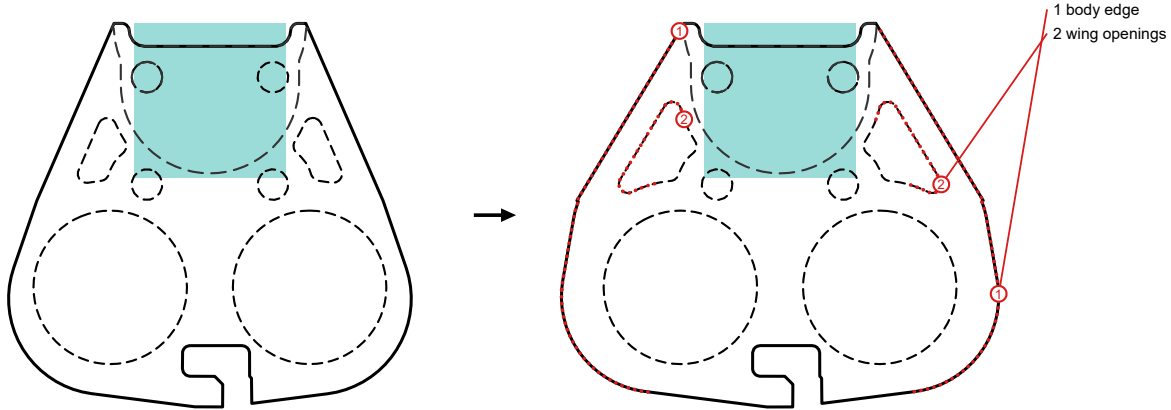
Before: 31.75. After: 41.75. Context: size = Custom.

Only acts when size is Custom.

Moves: body edge, wing openings.

custom_puller_middle_width

Body width at the middle



57.35 → 67.35 mm

Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom, a 25 mm wide plug in teal. Right: body width at the middle at 67.35 mm: widens the body at the middle control point, bulging or straightening the side edges between the widest point and the plug end. Marked in red: 1, the body edge: the outer outline. 2, the wing openings: the two openings beside the pocket. Everything else follows the outline.

Widens the body at the middle control point, bulging or straightening the side edges between the widest point and the plug end.

Default 57.35 · Range 25 to 120 · Step size 0.05 · Unit mm

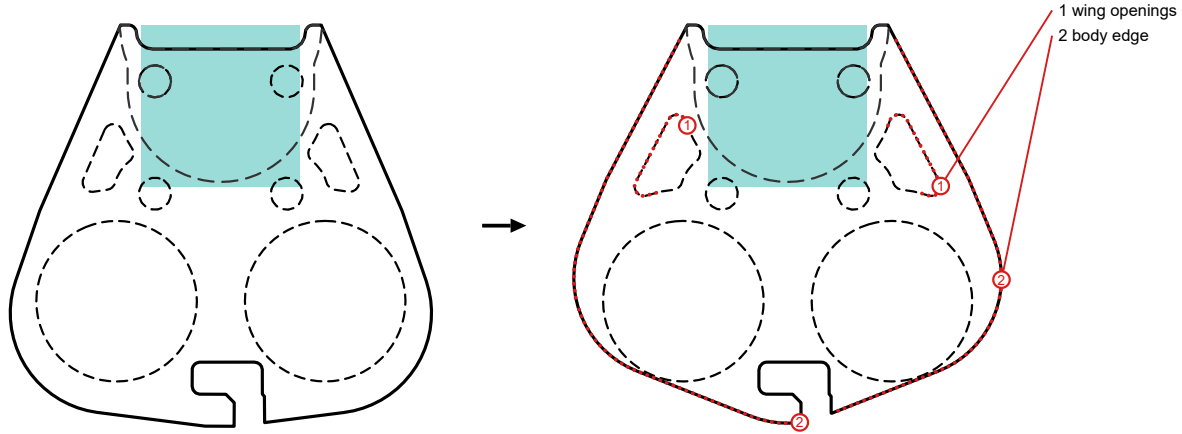
Before: 57.35. After: 67.35. Context: size = Custom.

Only acts when size is Custom.

Moves: body edge, wing openings.

custom_puller_side_corner

Side corner position



4.65 → 14.65 mm

Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom, a 25 mm wide plug in teal. Right: side corner position at 14.65 mm: moves the two widest corners of the body outline up toward the plug end or down toward the cord end. Marked in red: 1, the wing openings: the two openings beside the pocket. 2, the body edge: the outer outline. Everything else follows the outline.

Moves the two widest corners of the body outline up toward the plug end or down toward the cord end.

Default 4.65 · Range 3 to 35 · Step size 0.05 · Unit mm

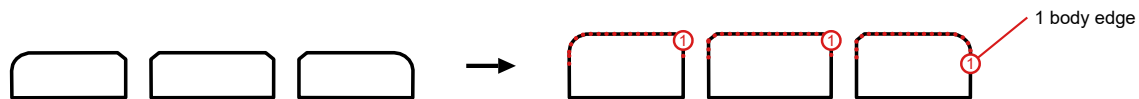
Before: 4.65. After: 14.65. Context: size = Custom.

Only acts when size is Custom.

Moves: body edge, wing openings.

custom_body_thickness

Body thickness



6.35 → 9 mm
section at y = 35 mm

Two vertical slices of the one-sided puller at y = 35 mm, before left and after right, the top face up. Left: the defaults with hand size set to Custom, the plug in teal in the cut. Right: body thickness at 9 mm: thickens the whole slab, so every hole and the pocket get deeper walls. Marked in red: 1, the body edge: the outer outline. Everything else follows the outline.

Thickens the whole slab, so every hole and the pocket get deeper walls.

Default 6.35 · Range 4 to 15 · Step size 0.05 · Unit mm

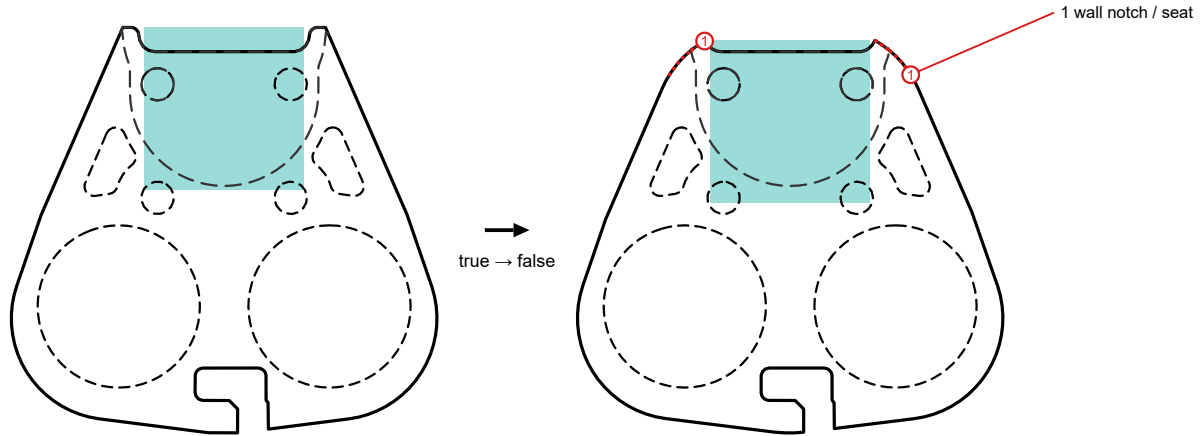
Before: 6.35. After: 9. Context: size = Custom. Section at y = 35 mm.

Only acts when size is Custom.

Moves: body edge.

custom_body_round_bottom_only

Round the cord half only



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom, a 25 mm wide plug in teal. Right: round the cord half only set to false: rounds the whole outline instead of only the cord half, so the plug end loses its crisp corners. Marked in red: 1, the wall notch and the seat: the notch in the top edge; the round recess at the plug end.

Rounds the whole outline instead of only the cord half, so the plug end loses its crisp corners.

Default true · true / false

Before: true. After: false. Context: size = Custom.

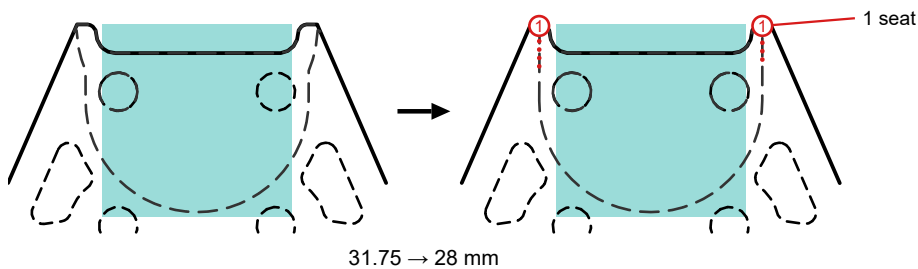
Only acts when size is Custom.

Moves: body edge, hook, seat, wall notch.

Plug Pocket (Custom size only)

custom_pocket_seat_diameter

Pocket seat diameter



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom, a 25 mm wide plug in teal. Right: pocket seat diameter at 28 mm: widens the round seat cut into the plug end of the pocket. Marked in red: 1, the seat: the round recess at the plug end. The body edge, the pocket, the wall notch and the wing openings stay where they were.

Widens the round seat cut into the plug end of the pocket.

Default 31.75 · Range 10 to 45 · Step size 0.05 · Unit mm

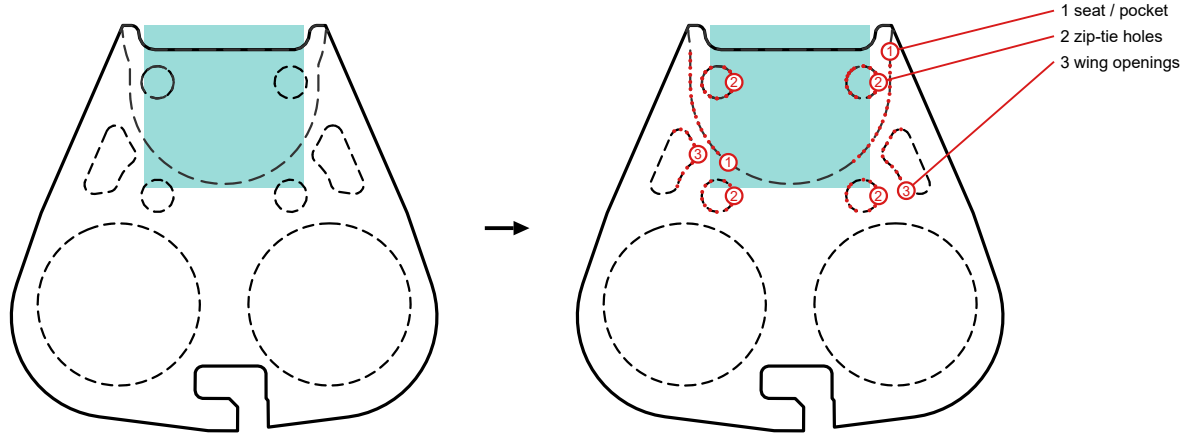
Before: 31.75. After: 28. Context: size = Custom.

Only acts when size is Custom.

Moves: seat.

custom_pocket_width

Pocket width



28.85 → 34 mm

Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom, a 25 mm wide plug in teal. Right: pocket width at 34 mm. Marked in red: 1, the seat and the pocket: the round recess at the plug end; the plug recess on the centerline. 2, the zip-tie holes: the four small holes beside the pocket. 3, the wing openings: the two openings beside the pocket.

Widens the plug recess, and the zip-tie holes and wing openings move outward with its wall.

Default 28.85 · Range 10 to 45 · Step size 0.05 · Unit mm

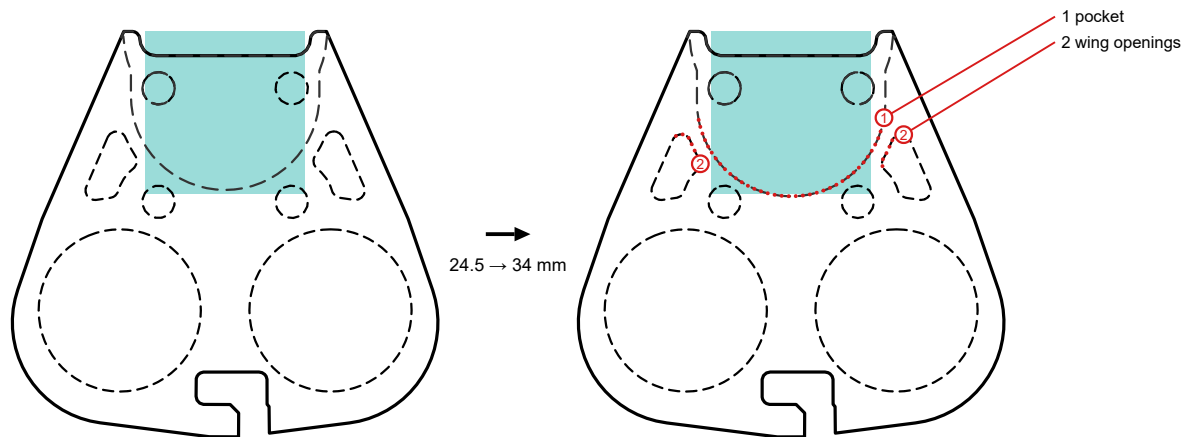
Before: 28.85. After: 34. Context: size = Custom.

Only acts when size is Custom.

Moves: pocket, seat, wing openings, zip-tie holes.

custom_pocket_depth

Pocket depth



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom, a 25 mm wide plug in teal. Right: pocket depth at 34 mm: runs the plug recess farther from the plug end toward the finger holes. Marked in red: 1, the pocket: the plug recess on the centerline. 2, the wing openings: the two openings beside the pocket.

Runs the plug recess farther from the plug end toward the finger holes.

Default 24.5 · Range 5 to 60 · Step size 0.5 · Unit mm

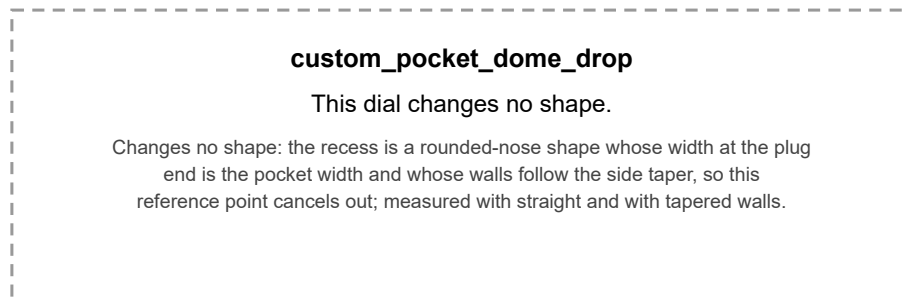
Before: 24.5. After: 34. Context: size = Custom.

Only acts when size is Custom.

Moves: pocket, wing openings.

custom_pocket_dome_drop

Pocket reference drop



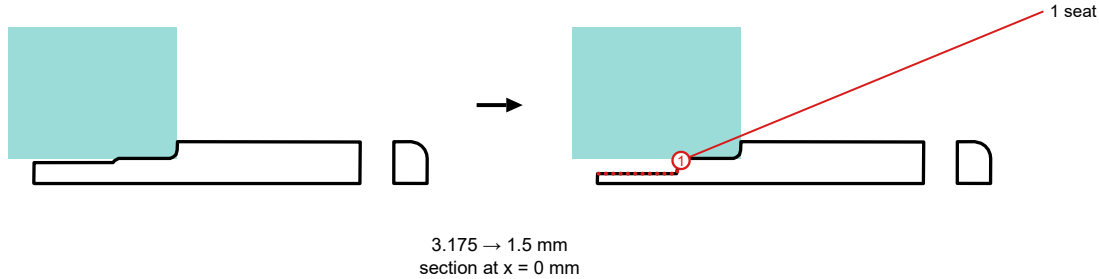
This dial changes no shape: the recess is a rounded-nose shape whose width at the plug end is the pocket width and whose walls follow the side taper, so this reference point cancels out; measured with straight and with tapered walls.

Changes no shape of the pocket, because the recess outline is built from the plug width at the plug end and the side taper, and this reference point cancels out of it.

Default 2.15 · Range 0 to 8 · Step size 0.05 · Unit mm

custom_pocket_seat_floor

Pocket seat floor thickness



Two vertical slices of the one-sided puller at $x = 0$ mm, before left and after right, the top face up. Left: the defaults with hand size set to Custom, the plug in teal in the cut. Right: pocket seat floor thickness at 1.5 mm: thins or thickens the floor left under the round seat, so the seat is cut deeper or shallower. Marked in red: 1, the seat: the round recess at the plug end.

Thins or thickens the floor left under the round seat, so the seat is cut deeper or shallower.

Default 3.175 · Range 0 to 8 · Step size 0.025 · Unit mm

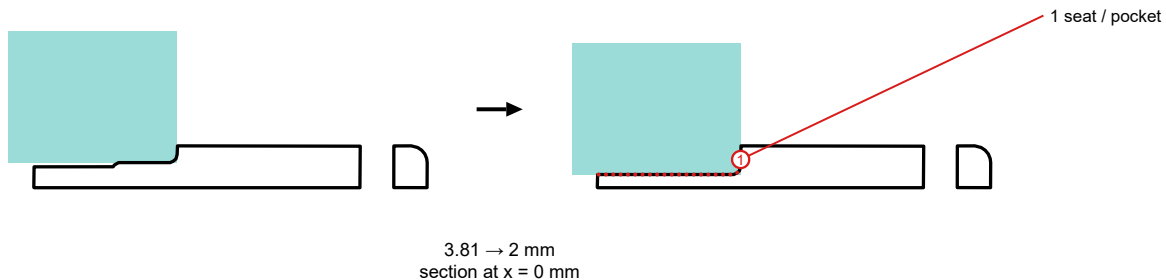
Before: 3.175. After: 1.5. Context: size = Custom. Section at $x = 0$ mm.

Only acts when size is Custom.

Moves: seat.

custom_pocket_floor

Plug recess floor thickness



Two vertical slices of the one-sided puller at $x = 0$ mm, before left and after right, the top face up. Left: the defaults with hand size set to Custom, the plug in teal in the cut. Right: plug recess floor thickness at 2 mm: thins or thickens the floor left under the plug recess, so the recess is cut deeper or shallower. Marked in red: 1, the seat and the pocket: the round recess at the plug end; the plug recess on the centerline.

Thins or thickens the floor left under the plug recess, so the recess is cut deeper or shallower.

Default 3.81 · Range 0 to 12 · Step size 0.025 · Unit mm

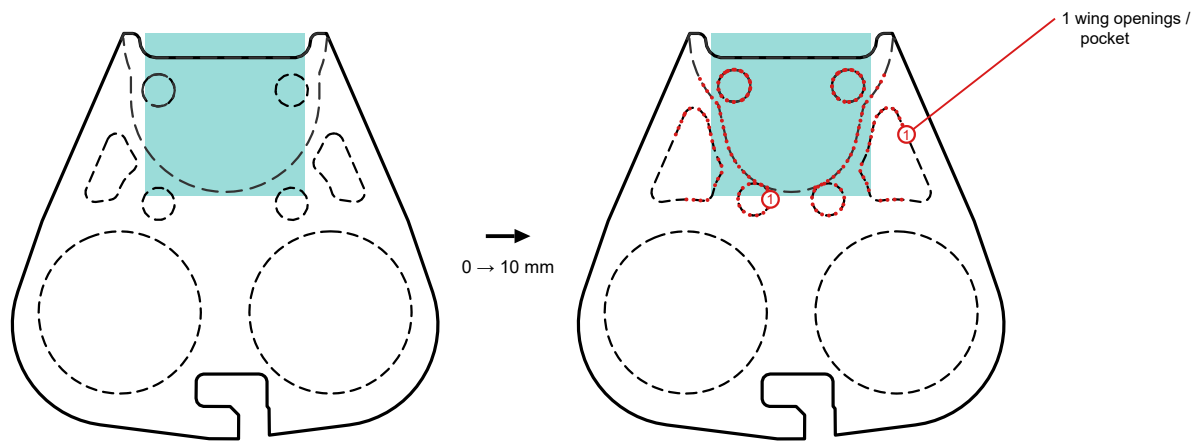
Before: 3.81. After: 2. Context: size = Custom. Section at $x = 0$ mm.

Only acts when size is Custom.

Moves: pocket, seat.

custom_pocket_side_angle

Pocket side taper



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom, a 25 mm wide plug in teal. Right: pocket side taper at 10 mm: tilts the pocket's side walls inward toward the cord end, and the zip-tie holes and strap slots lean in along the same line. Marked in red: 1, the wing openings and the pocket: the two openings beside the pocket; the plug recess on the centerline.

Tilts the pocket's side walls inward toward the cord end, and the zip-tie holes and strap slots lean in along the same line.

Default 0 · Range -15 to 25 · Step size 0.5 · Unit mm

Before: 0. After: 10. Context: size = Custom.

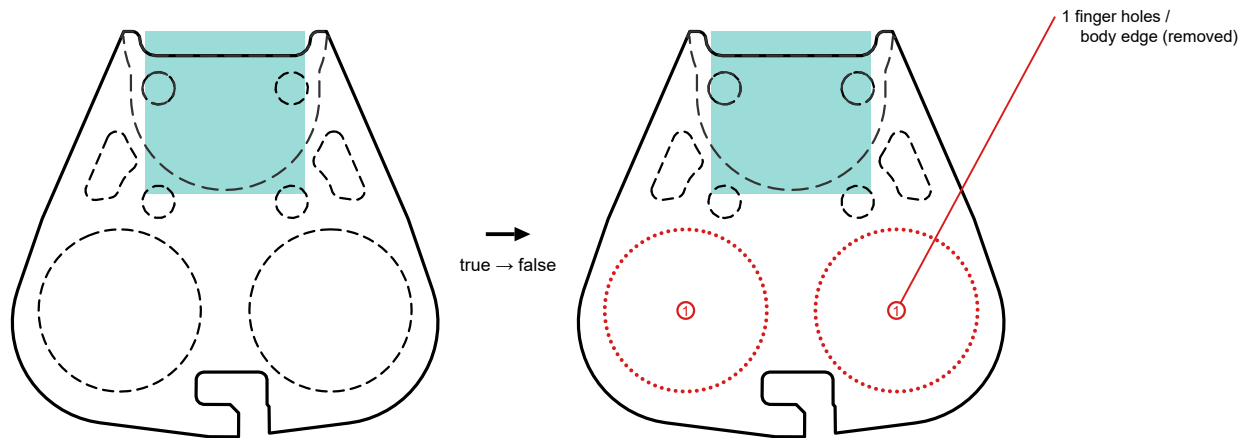
Only acts when size is Custom.

Moves: pocket, wing openings.

Finger Holes (Custom size only)

`custom_enable_finger_holes`

Finger holes on or off



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom, a 25 mm wide plug in teal. Right: finger holes on or off set to false: cuts or leaves out both finger holes. Marked in red: 1, the finger holes and the body edge; the two large holes in the lower half; the outer outline, removed. Everything else follows the outline.

Cuts or leaves out both finger holes.

Default true · true / false

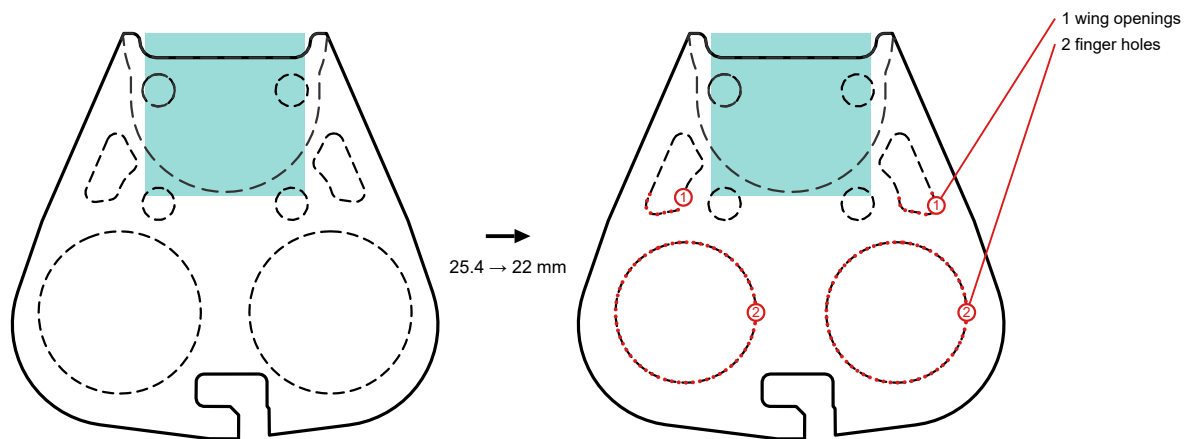
Before: true. After: false. Context: size = Custom.

Only acts when size is Custom.

Moves: body edge, finger holes.

custom_finger_hole_diameter

Finger hole diameter



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom, a 25 mm wide plug in teal. Right: finger hole diameter at 22 mm: widens both finger holes. Marked in red: 1, the wing openings: the two openings beside the pocket. 2, the finger holes: the two large holes in the lower half. The body edge, the pocket, the seat and the wall notch stay where they were.

Widens both finger holes.

Default 25.4 · Range 15 to 40 · Step size 0.1 · Unit mm

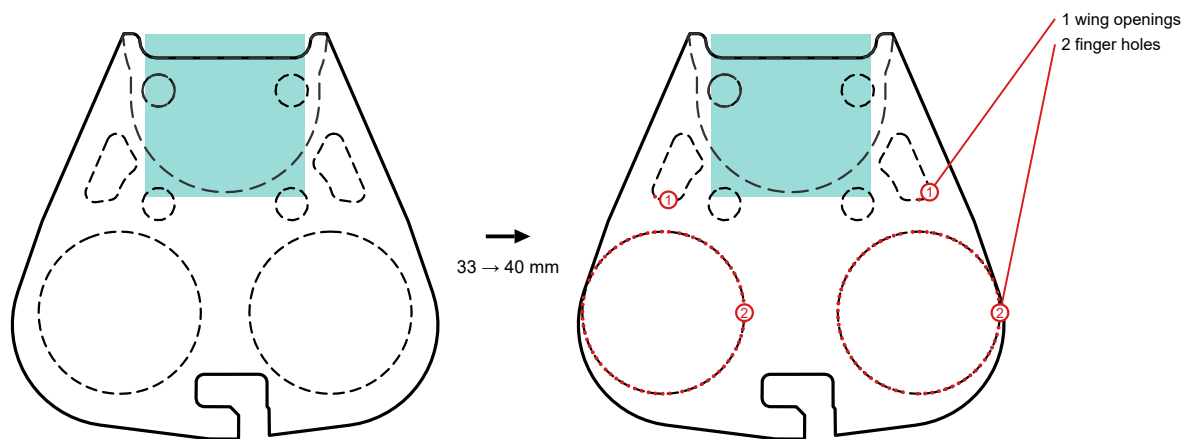
Before: 25.4. After: 22. Context: size = Custom.

Only acts when size is Custom.

Moves: finger holes, wing openings.

custom_finger_hole_spacing

Finger hole spacing



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom, a 25 mm wide plug in teal. Right: finger hole spacing at 40 mm: moves the two finger holes farther apart or closer together. Marked in red: 1, the wing openings: the two openings beside the pocket. 2, the finger holes: the two large holes in the lower half.

Moves the two finger holes farther apart or closer together.

Default 33 · Range 20 to 50 · Step size 0.5 · Unit mm

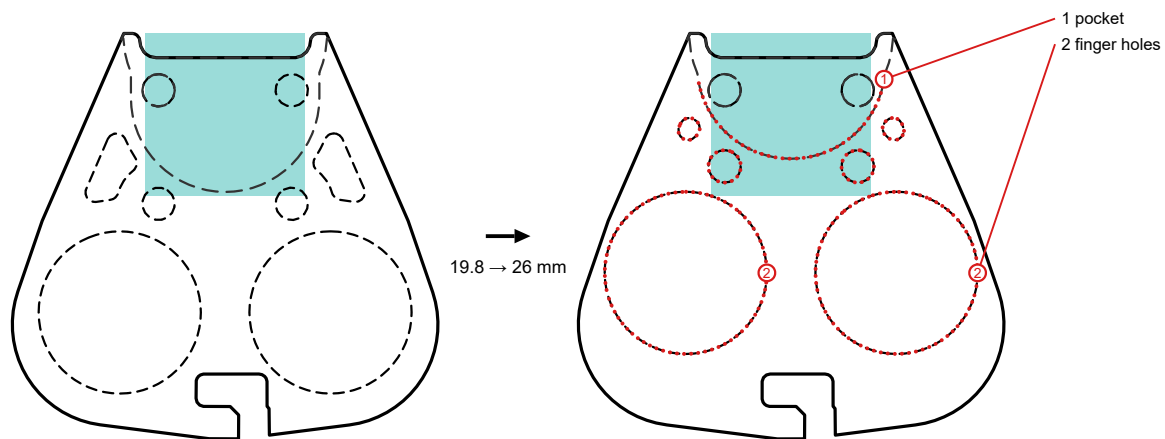
Before: 33. After: 40. Context: size = Custom.

Only acts when size is Custom.

Moves: finger holes, wing openings.

custom_finger_hole_y_position

Finger hole position



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom, a 25 mm wide plug in teal. Right: finger hole position at 26 mm: slides both finger holes up toward the pocket or down toward the cord hook. Marked in red: 1, the pocket: the plug recess on the centerline. 2, the finger holes: the two large holes in the lower half.

Slides both finger holes up toward the pocket or down toward the cord hook.

Default 19.8 · Range 10 to 50 · Step size 0.1 · Unit mm

Before: 19.8. After: 26. Context: size = Custom.

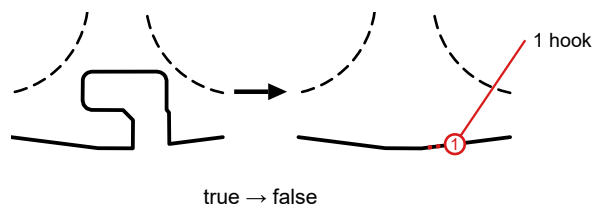
Only acts when size is Custom.

Moves: finger holes, pocket.

T Hook (Custom size only)

custom_enable_t_hook

Cord hook on or off



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom, a 25 mm wide plug in teal. Right: cord hook on or off set to false: cuts or leaves out the cord hook at the cord end. Marked in red: 1, the hook: the cord hook slot in the bottom edge. The body edge, the pocket, the seat and the wall notch stay where they were.

Cuts or leaves out the cord hook at the cord end.

Default true · true / false

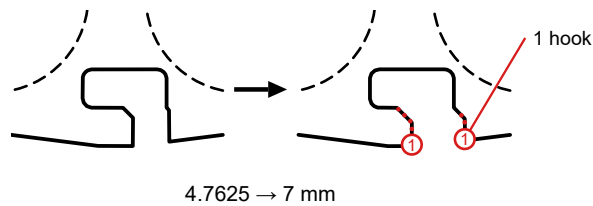
Before: true. After: false. Context: size = Custom.

Only acts when size is Custom.

Moves: hook.

custom_t_hook_base_gap

Hook slot width



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom, a 25 mm wide plug in teal. Right: hook slot width at 7 mm: widens the slot the cord slides into at the cord end. Marked in red: 1, the hook: the cord hook slot in the bottom edge. The body edge, the pocket, the seat and the wall notch stay where they were.

Widens the slot the cord slides into at the cord end.

Default 4.7625 · Range 3 to 10 · Step size 0.05 · Unit mm

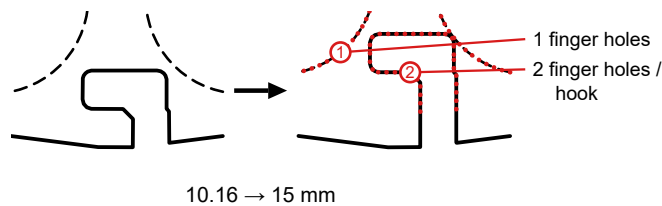
Before: 4.7625. After: 7. Context: size = Custom.

Only acts when size is Custom.

Moves: hook.

custom_t_hook_length

Hook length



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom, a 25 mm wide plug in teal. Right: hook length at 15 mm: runs the cord hook farther into the body from the cord end. Marked in red: 1, the finger holes: the two large holes in the lower half. 2, the finger holes and the hook: the two large holes in the lower half; the cord hook slot in the bottom edge.

Runs the cord hook farther into the body from the cord end.

Default 10.16 · Range 6 to 20 · Step size 0.05 · Unit mm

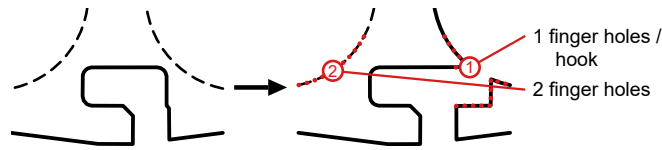
Before: 10.16. After: 15. Context: size = Custom.

Only acts when size is Custom.

Moves: finger holes, hook, pocket, wing openings, zip-tie holes.

custom_t_hook_holder_width

Hook crossbar width



11.1125 → 16 mm

Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom, a 25 mm wide plug in teal. Right: hook crossbar width at 16 mm. Marked in red: 1, the finger holes and the hook: the two large holes in the lower half; the cord hook slot in the bottom edge. 2, the finger holes: the two large holes in the lower half.

Widens the crossbar the cord hooks under at the top of the slot.

Default 11.1125 · Range 8 to 25 · Step size 0.05 · Unit mm

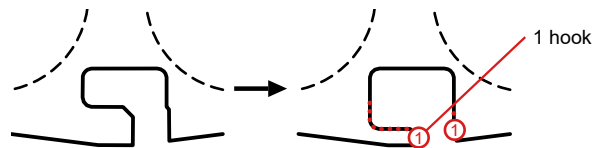
Before: 11.1125. After: 16. Context: size = Custom.

Only acts when size is Custom.

Moves: finger holes, hook, wing openings, zip-tie holes.

custom_t_hook_holder_length

Hook crossbar height



5.08 → 8 mm

Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom, a 25 mm wide plug in teal. Right: hook crossbar height at 8 mm: makes the crossbar opening taller or shorter along the body. Marked in red: 1, the hook: the cord hook slot in the bottom edge. The body edge, the pocket, the seat and the wall notch stay where they were.

Makes the crossbar opening taller or shorter along the body.

Default 5.08 · Range 2 to 15 · Step size 0.05 · Unit mm

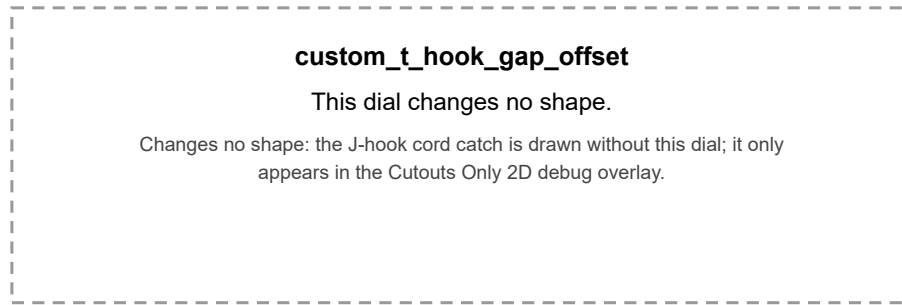
Before: 5.08. After: 8. Context: size = Custom.

Only acts when size is Custom.

Moves: hook.

custom_t_hook_gap_offset

Hook slot inset



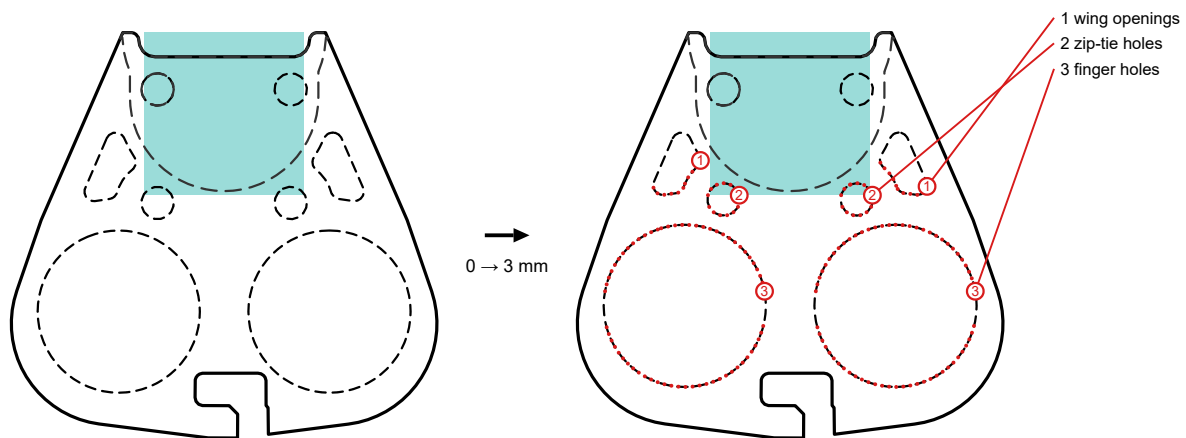
This dial changes no shape: the J-hook cord catch is drawn without this dial; it only appears in the Cutouts Only 2D debug overlay.

Changes no shape of the printed tool, because the J-hook cord catch does not read this dial.

Default 0 · Range 0 to 8 · Step size 0.5 · Unit mm

custom_t_hook_leg_offset

Hook stem side offset



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom, a 25 mm wide plug in teal. Right: hook stem side offset at 3 mm. Marked in red: 1, the wing openings: the two openings beside the pocket. 2, the zip-tie holes: the four small holes beside the pocket. 3, the finger holes: the two large holes in the lower half.

Moves nothing on the hook itself, and only pushes the finger holes up when the auto-fit keep-out around the hook grows with it.

Default 0 · Range -5 to 5 · Step size 0.5 · Unit mm

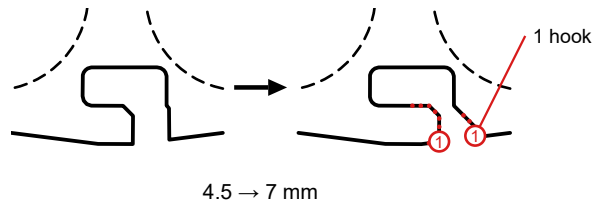
Before: 0. After: 3. Context: size = Custom.

Only acts when size is Custom; the J-hook cord catch is drawn without this dial, and only the auto-fit clamp on the finger holes reads it.

Moves: finger holes, wing openings, zip-tie holes.

custom_t_hook_stem_offset

Hook stem offset



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom, a 25 mm wide plug in teal. Right: hook stem offset at 7 mm: shifts the cord slot sideways along the crossbar, making the J of the hook deeper or shallower. Marked in red: 1, the hook: the cord hook slot in the bottom edge. The body edge, the pocket, the seat and the wall notch stay where they were.

Shifts the cord slot sideways along the crossbar, making the J of the hook deeper or shallower.

Default 4.5 · Range 0 to 8 · Step size 0.05 · Unit mm

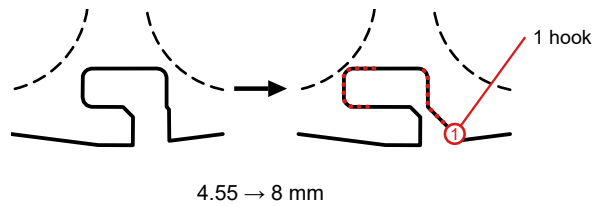
Before: 4.5. After: 7. Context: size = Custom.

Only acts when size is Custom.

Moves: hook.

custom_t_hook_catch_reach

Hook catch reach



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom, a 25 mm wide plug in teal. Right: hook catch reach at 8 mm: extends the crossbar past the slot on the catch side, lengthening the lip the cord hooks under. Marked in red: 1, the hook: the cord hook slot in the bottom edge. The body edge, the pocket, the seat and the wall notch stay where they were.

Extends the crossbar past the slot on the catch side, lengthening the lip the cord hooks under.

Default 4.55 · Range 0 to 10 · Step size 0.05 · Unit mm

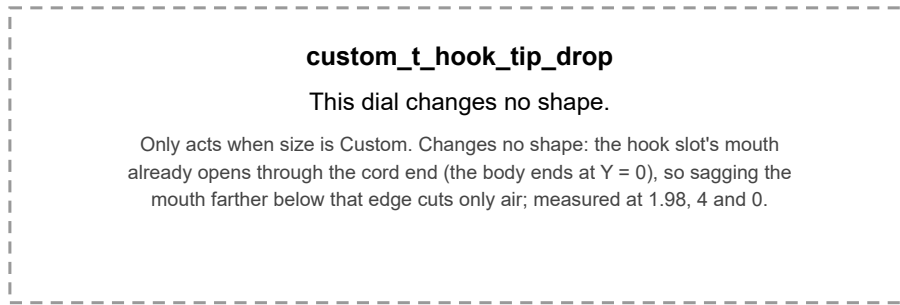
Before: 4.55. After: 8. Context: size = Custom.

Only acts when size is Custom.

Moves: hook.

custom_t_hook_tip_drop

Hook tip drop



This dial changes no shape. Only acts when size is Custom. Changes no shape: the hook slot's mouth already opens through the cord end (the body ends at Y = 0), so sagging the mouth farther below that edge cuts only air; measured at 1.98, 4 and 0.

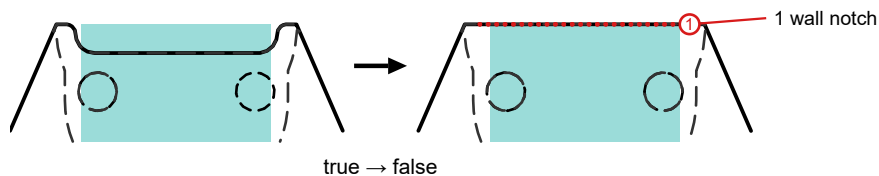
Sags the slot's mouth farther below the cord end so a hooked cord cannot back out.

Default 1.98 · Range 0 to 5 · Step size 0.05 · Unit mm

Plug Wall Notch (Custom size only)

custom_enable_plug_wall_notch

Wall notch on or off



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom, a 25 mm wide plug in teal. Right: wall notch on or off set to false: cuts or leaves out the notch at the plug end that straddles the outlet's cover plate. Marked in red: 1, the wall notch: the notch in the top edge. The body edge, the pocket, the seat and the wing openings stay where they were.

Cuts or leaves out the notch at the plug end that straddles the outlet's cover plate.

Default true · true / false

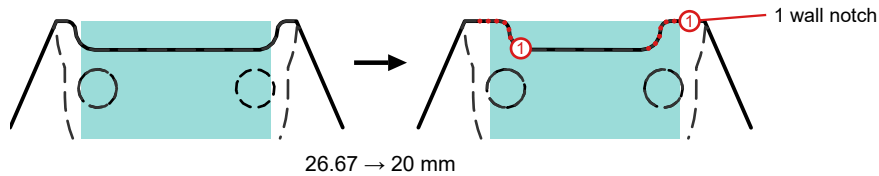
Before: true. After: false. Context: size = Custom.

Only acts when size is Custom.

Moves: wall notch.

custom_plug_wall_notch_width

Wall notch width



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom, a 25 mm wide plug in teal. Right: wall notch width at 20 mm: widens or narrows the notch at the plug end. Marked in red: 1, the wall notch: the notch in the top edge. The body edge, the pocket, the seat and the wing openings stay where they were.

Widens or narrows the notch at the plug end.

Default 26.67 · Range 5 to 40 · Step size 0.01 · Unit mm

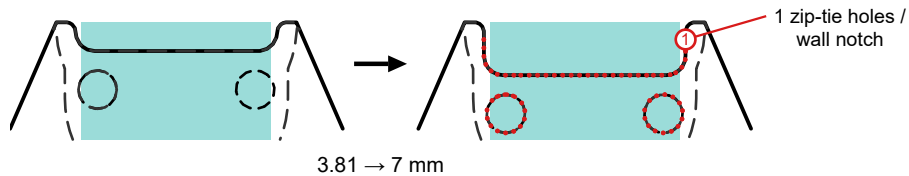
Before: 26.67. After: 20. Context: size = Custom.

Only acts when size is Custom.

Moves: wall notch.

custom_plug_wall_notch_height

Wall notch depth



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom, a 25 mm wide plug in teal. Right: wall notch depth at 7 mm: cuts the notch deeper or shallower into the plug end. Marked in red: 1, the zip-tie holes and the wall notch: the four small holes beside the pocket; the notch in the top edge. The body edge, the pocket, the seat and the wing openings stay where they were.

Cuts the notch deeper or shallower into the plug end.

Default 3.81 · Range 0 to 10 · Step size 0.05 · Unit mm

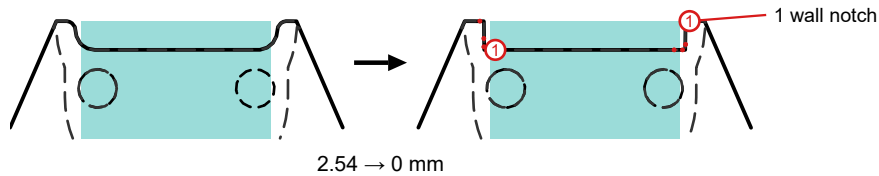
Before: 3.81. After: 7. Context: size = Custom.

Only acts when size is Custom.

Moves: wall notch, wing openings, zip-tie holes.

custom_plug_wall_notch_rounding

Wall notch corner rounding



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom, a 25 mm wide plug in teal. Right: wall notch corner rounding at 0 mm: rounds or squares the notch's two inner corners. Marked in red: 1, the wall notch: the notch in the top edge. The body edge, the pocket, the seat and the wing openings stay where they were.

Rounds or squares the notch's two inner corners.

Default 2.54 · Range 0 to 5 · Step size 0.01 · Unit mm

Before: 2.54. After: 0. Context: size = Custom.

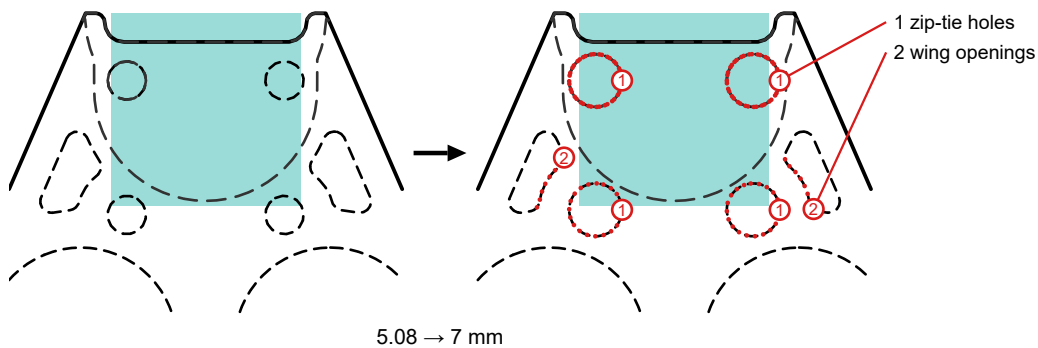
Only acts when size is Custom.

Moves: wall notch.

Zip Tie Holes (Custom size only)

custom_zip_tie_hole_diameter

Zip-tie hole diameter



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom, a 25 mm wide plug in teal. Right: zip-tie hole diameter at 7 mm: widens every zip-tie hole. Marked in red: 1, the zip-tie holes: the four small holes beside the pocket. 2, the wing openings: the two openings beside the pocket. The body edge, the pocket, the seat and the wall notch stay where they were.

Widens every zip-tie hole.

Default 5.08 · Range 2 to 8 · Step size 0.02 · Unit mm

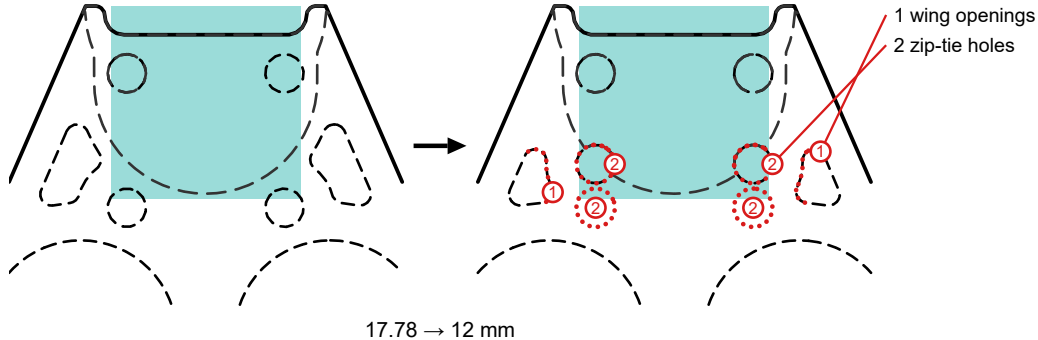
Before: 5.08. After: 7. Context: size = Custom.

Only acts when size is Custom.

Moves: wing openings, zip-tie holes.

custom_zip_tie_height_spacing

Zip-tie row spacing



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom, a 25 mm wide plug in teal. Right: zip-tie row spacing at 12 mm: moves the second row of zip-tie holes closer to or farther from the first. Marked in red: 1, the wing openings: the two openings beside the pocket. 2, the zip-tie holes: the four small holes beside the pocket.

Moves the second row of zip-tie holes closer to or farther from the first.

Default 17.78 · Range 5 to 30 · Step size 0.02 · Unit mm

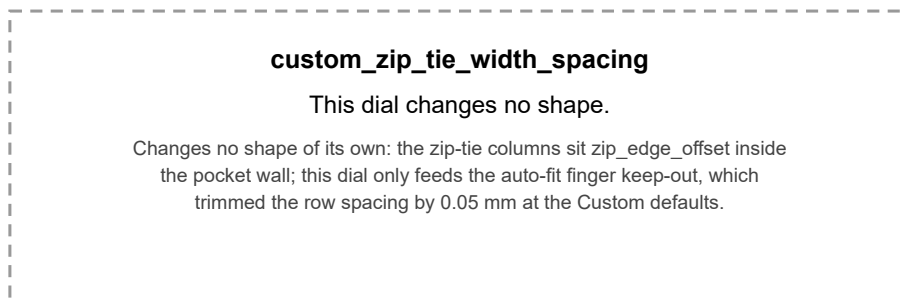
Before: 17.78. After: 12. Context: size = Custom.

Only acts when size is Custom.

Moves: wing openings, zip-tie holes.

custom_zip_tie_width_spacing

Zip-tie column spacing



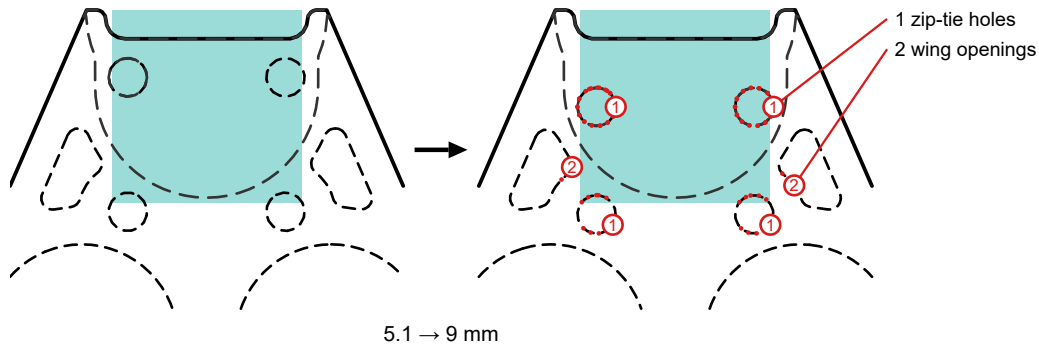
This dial changes no shape. Changes no shape of its own: the zip-tie columns sit `zip_edge_offset` inside the pocket wall; this dial only feeds the auto-fit finger keep-out, which trimmed the row spacing by 0.05 mm at the Custom defaults.

Moves no zip-tie hole, because the two columns follow the pocket wall, and only feeds the auto-fit keep-out that can trim the row spacing by a hair.

Default 17.7 · Range 10 to 50 · Step size 0.02 · Unit mm

custom_zip_tie_distance_from_notch

Zip-tie distance from the notch



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom, a 25 mm wide plug in teal. Right: zip-tie distance from the notch at 9 mm: moves the first row of zip-tie holes farther from the wall notch, and the second row follows. Marked in red: 1, the zip-tie holes: the four small holes beside the pocket. 2, the wing openings: the two openings beside the pocket.

Moves the first row of zip-tie holes farther from the wall notch, and the second row follows.

Default 5.1 · Range 1 to 15 · Step size 0.05 · Unit mm

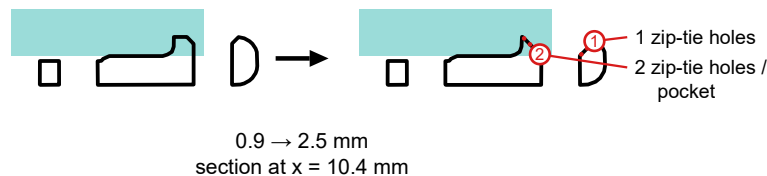
Before: 5.1. After: 9. Context: size = Custom.

Only acts when size is Custom.

Moves: wing openings, zip-tie holes.

custom_zip_tie_countersink

Zip-tie countersink



Two vertical slices of the one-sided puller at x = 10.4 mm, before left and after right, the top face up. Left: the defaults with hand size set to Custom, the plug in teal in the cut. Right: zip-tie countersink at 2.5 mm. Marked in red: 1, the zip-tie holes: the four small holes beside the pocket. 2, the zip-tie holes and the pocket: the four small holes beside the pocket; the plug recess on the centerline.

Flares the top of the zip-tie holes wider so the tie head sits flush.

Default 0.9 · Range 0 to 3 · Step size 0.05 · Unit mm

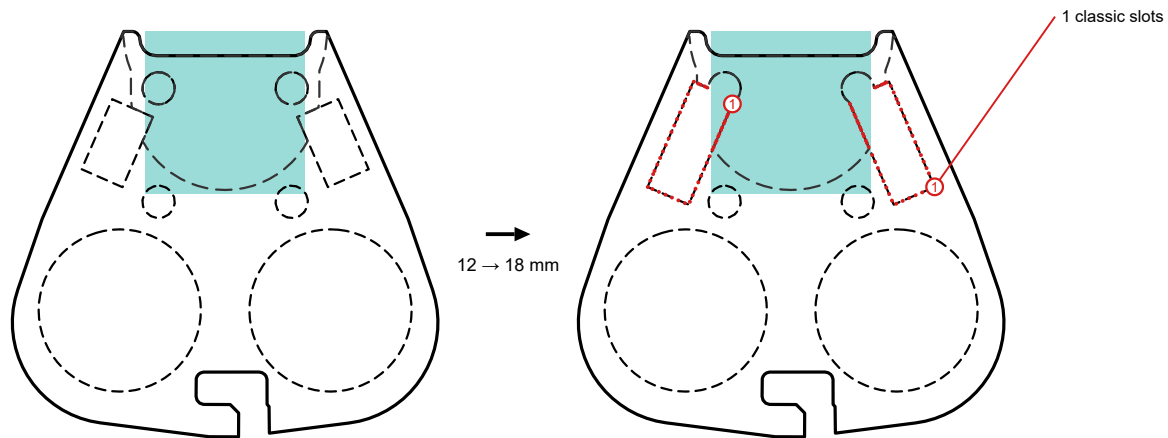
Before: 0.9. After: 2.5. Context: size = Custom. Section at x = 10.4 mm.

Only acts when size is Custom.

Moves: pocket, zip-tie holes.

Velcro / Wing Strap Holes (Custom size only)

custom_velcro_hole_length
Strap slot length



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom and strap opening style set to Classic slot, a 25 mm wide plug in teal. Right: strap slot length at 18 mm: lengthens each plain strap slot along its lean. Marked in red: 1, the classic slots: the two slots beside the pocket. The body edge, the pocket, the seat and the wall notch stay where they were.

Lengthens each plain strap slot along its lean.

Default 12 · Range 6 to 20 · Step size 0.5 · Unit mm

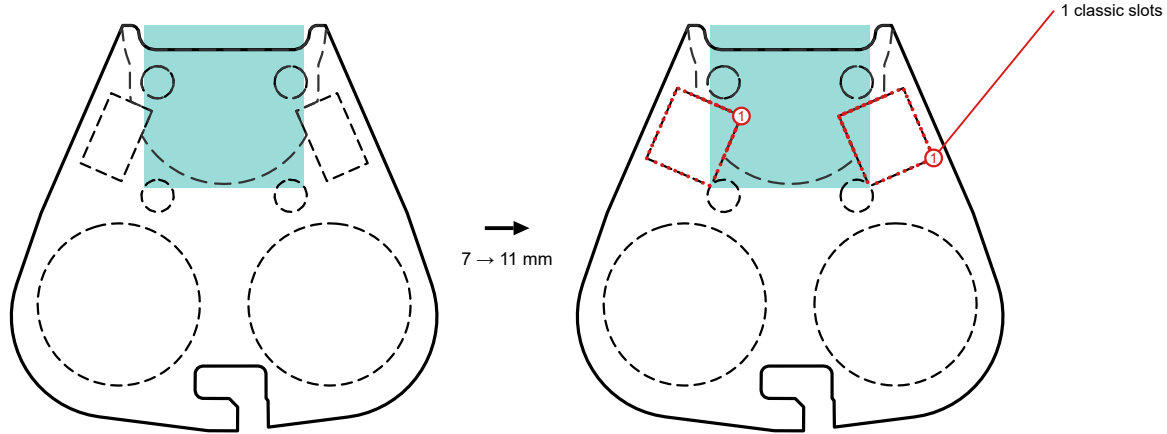
Before: 12. After: 18. Context: size = Custom, velcro_style = Classic slot.

Only acts when size is Custom and velcro_style is Classic slot.

Moves: classic slots.

custom_velcro_hole_width

Strap slot width



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom and strap opening style set to Classic slot, a 25 mm wide plug in teal. Right: strap slot width at 11 mm: widens each plain strap slot. Marked in red: 1, the classic slots: the two slots beside the pocket. The body edge, the pocket, the seat and the wall notch stay where they were.

Widens each plain strap slot.

Default 7 · Range 3 to 14 · Step size 0.5 · Unit mm

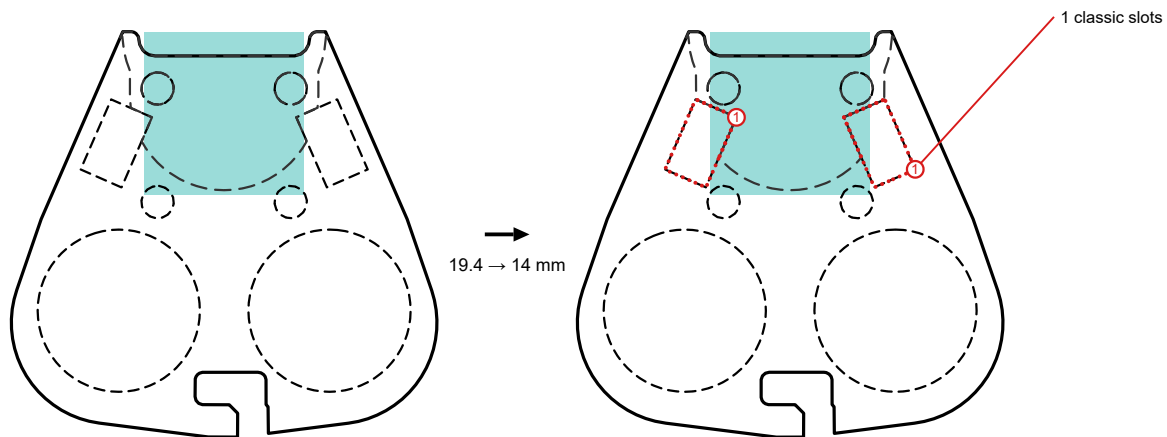
Before: 7. After: 11. Context: size = Custom, velcro_style = Classic slot.

Only acts when size is Custom and velcro_style is Classic slot.

Moves: classic slots.

custom_velcro_hole_x_center

Strap slot distance from the centerline



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom and strap opening style set to Classic slot, a 25 mm wide plug in teal. Right: strap slot distance from the centerline at 14 mm: moves the two plain strap slots closer to or farther from the centerline. Marked in red: 1, the classic slots: the two slots beside the pocket.

Moves the two plain strap slots closer to or farther from the centerline.

Default 19.4 · Range 5 to 35 · Step size 0.05 · Unit mm

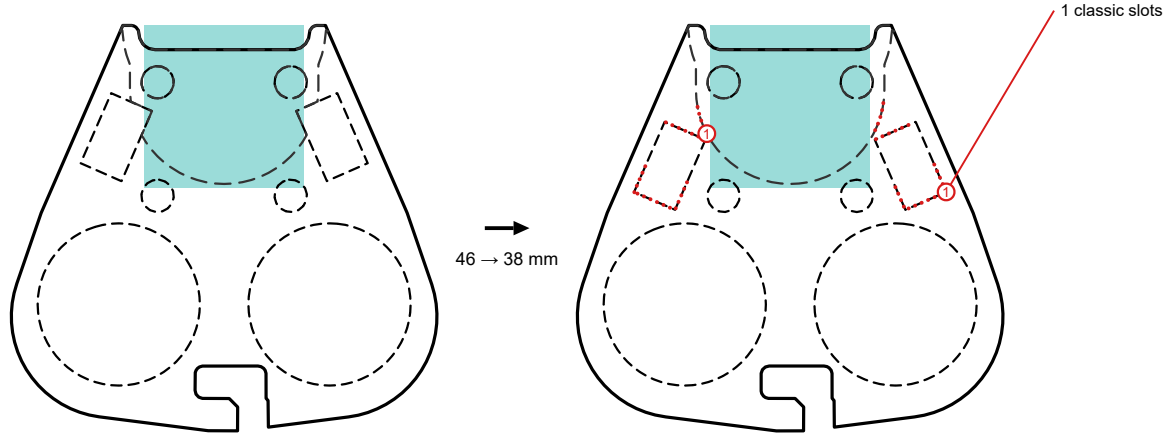
Before: 19.4. After: 14. Context: size = Custom, velcro_style = Classic slot.

Only acts when size is Custom and velcro_style is Classic slot.

Moves: classic slots.

custom_velcro_hole_y_center

Strap slot position along the body



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom and strap opening style set to Classic slot, a 25 mm wide plug in teal. Right: strap slot position along the body at 38 mm: slides the two plain strap slots toward the cord end or the plug end. Marked in red: 1, the classic slots: the two slots beside the pocket.

Slides the two plain strap slots toward the cord end or the plug end.

Default 46 · Range 30 to 80 · Step size 0.5 · Unit mm

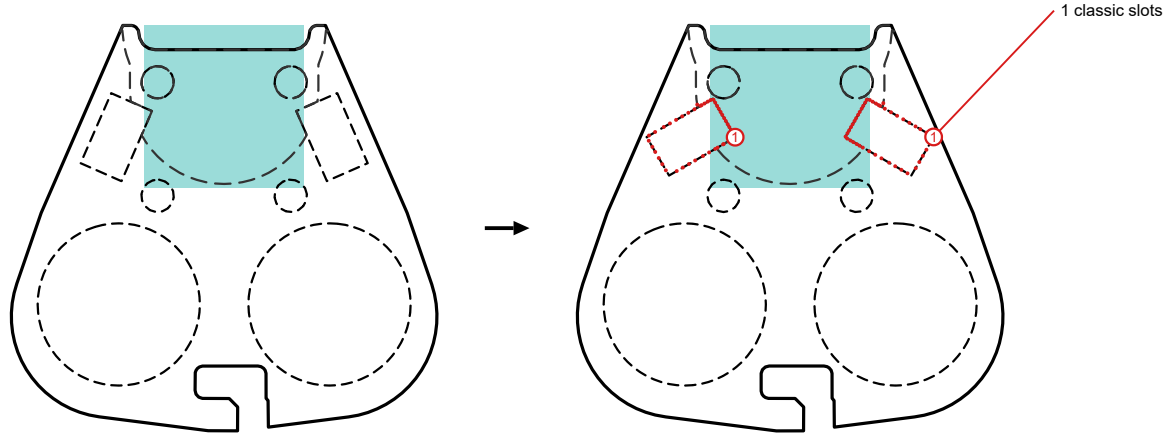
Before: 46. After: 38. Context: size = Custom, velcro_style = Classic slot.

Only acts when size is Custom and velcro_style is Classic slot.

Moves: classic slots.

custom_velcro_hole_rotation

Strap slot lean



23.5 → 60 deg

Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom and strap opening style set to Classic slot, a 25 mm wide plug in teal. Right: strap slot lean at 60 deg: leans the two plain strap slots more or less steeply, mirrored on the two sides. Marked in red: 1, the classic slots: the two slots beside the pocket.

Leans the two plain strap slots more or less steeply, mirrored on the two sides.

Default 23.5 · Range 0 to 180 · Step size 0.5 · Unit deg

Before: 23.5. After: 60. Context: size = Custom, velcro_style = Classic slot.

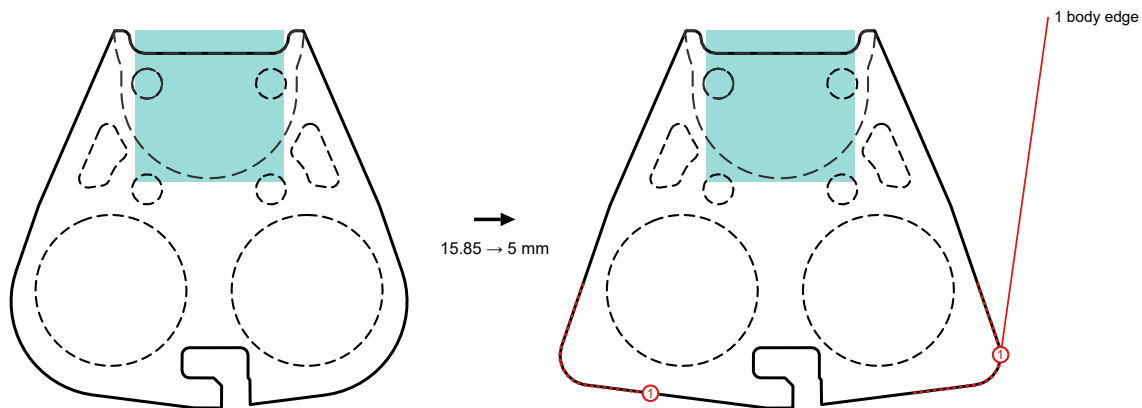
Only acts when size is Custom and velcro_style is Classic slot.

Moves: classic slots.

Edge Rounding (Custom size only)

custom_body_side_rounding

Body side rounding



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom, a 25 mm wide plug in teal. Right: body side rounding at 5 mm: rounds the body outline's corners more or less, down to the plain octagon at 0. Marked in red: 1, the body edge: the outer outline. Everything else follows the outline.

Rounds the body outline's corners more or less, down to the plain octagon at 0.

Default 15.85 · Range 0 to 30 · Step size 0.05 · Unit mm

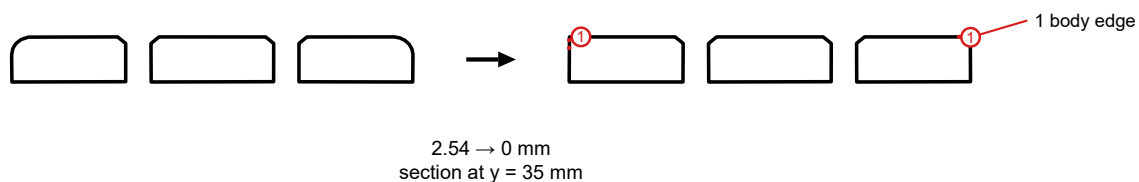
Before: 15.85. After: 5. Context: size = Custom.

Only acts when size is Custom.

Moves: body edge.

custom_body_top_rounding

Top edge rounding



Two vertical slices of the one-sided puller at y = 35 mm, before left and after right, the top face up. Left: the defaults with hand size set to Custom, the plug in teal in the cut. Right: top edge rounding at 0 mm: rounds or squares the edge where the body's top face meets its sides. Marked in red: 1, the body edge: the outer outline. Everything else follows the outline.

Rounds or squares the edge where the body's top face meets its sides.

Default 2.54 · Range 0 to 5 · Step size 0.01 · Unit mm

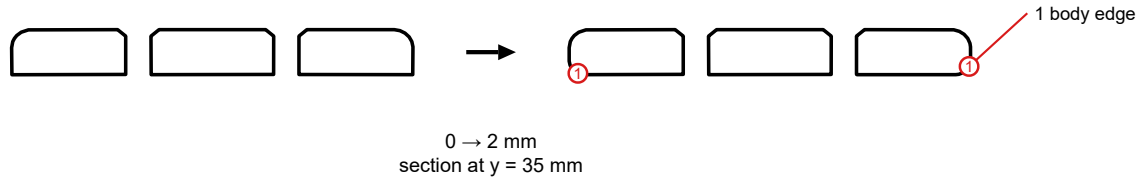
Before: 2.54. After: 0. Context: size = Custom. Section at y = 35 mm.

Only acts when size is Custom.

Moves: body edge.

custom_body_bottom_rounding

Bottom edge rounding



Two vertical slices of the one-sided puller at $y = 35$ mm, before left and after right, the top face up. Left: the defaults with hand size set to Custom, the plug in teal in the cut. Right: bottom edge rounding at 2 mm: rounds or squares the edge where the body's bottom face meets its sides. Marked in red: 1, the body edge: the outer outline. Everything else follows the outline.

Rounds or squares the edge where the body's bottom face meets its sides.

Default 0 · Range 0 to 5 · Step size 0.1 · Unit mm

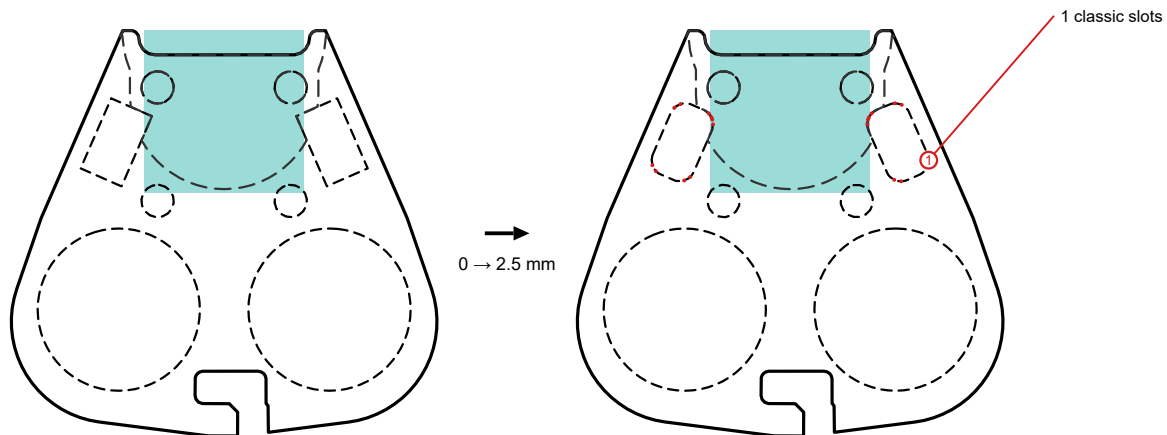
Before: 0. After: 2. Context: size = Custom. Section at $y = 35$ mm.

Only acts when size is Custom.

Moves: body edge.

custom_velcro_side_rounding

Strap slot corner rounding



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom and strap opening style set to Classic slot, a 25 mm wide plug in teal. Right: strap slot corner rounding at 2.5 mm: rounds the four corners of each plain strap slot. Marked in red: 1, the classic slots: the two slots beside the pocket. The body edge, the pocket, the seat and the wall notch stay where they were.

Rounds the four corners of each plain strap slot.

Default 0 · Range 0 to 3 · Step size 0.5 · Unit mm

Before: 0. After: 2.5. Context: size = Custom, velcro_style = Classic slot.

Only acts when size is Custom and velcro_style is Classic slot.

Moves: classic slots.

custom_velcro_top_bottom_rounding

Strap opening edge rounding



0 → 1.5 mm
section at y = 46 mm

Two vertical slices of the one-sided puller at y = 46 mm, before left and after right, the top face up. Left: the defaults with hand size set to Custom, the plug in teal in the cut. Right: strap opening edge rounding at 1.5 mm. Marked in red: 1, the wing openings: the two openings beside the pocket. 2, the wing openings and the pocket: the two openings beside the pocket; the plug recess on the centerline.

Flares the top and bottom edges of the strap openings so the strap does not chafe.

Default 0 · Range 0 to 3 · Step size 0.1 · Unit mm

Before: 0. After: 1.5. Context: size = Custom. Section at y = 46 mm.

Only acts when size is Custom.

Moves: pocket, wing openings.

custom_finger_hole_rounding

Finger hole rim rounding



2.5 → 0 mm
section at y = 19.8 mm

Two vertical slices of the one-sided puller at y = 19.8 mm, before left and after right, the top face up. Left: the defaults with hand size set to Custom, the plug in teal in the cut. Right: finger hole rim rounding at 0 mm: rounds or squares the rims of both finger holes on both faces. Marked in red: 1, the finger holes: the two large holes in the lower half. The body edge, the pocket, the seat and the wall notch stay where they were.

Rounds or squares the rims of both finger holes on both faces.

Default 2.5 · Range 0 to 5 · Step size 0.1 · Unit mm

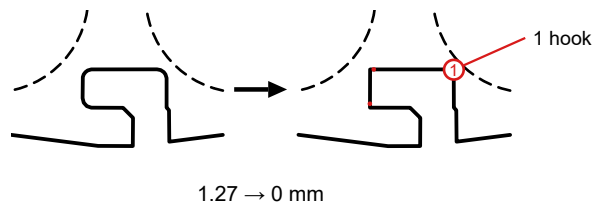
Before: 2.5. After: 0. Context: size = Custom. Section at y = 19.8 mm.

Only acts when size is Custom.

Moves: finger holes.

custom_t_hook_holder_side_rounding

Hook crossbar corner rounding



Two top views of the one-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Custom, a 25 mm wide plug in teal. Right: hook crossbar corner rounding at 0 mm: rounds or squares the top corners of the hook's crossbar. Marked in red: 1, the hook: the cord hook slot in the bottom edge. The body edge, the pocket, the seat and the wall notch stay where they were.

Rounds or squares the top corners of the hook's crossbar.

Default 1.27 · Range 0 to 3 · Step size 0.01 · Unit mm

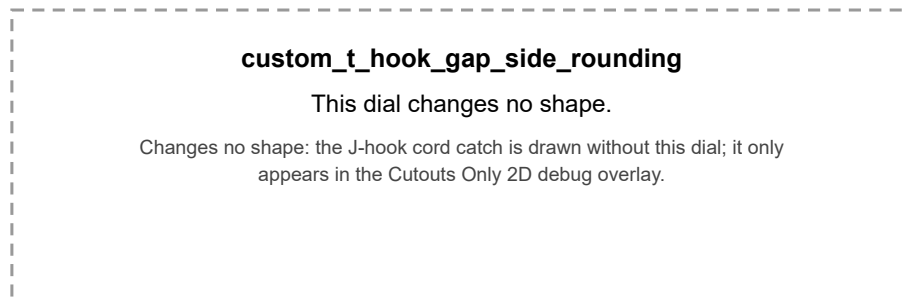
Before: 1.27. After: 0. Context: size = Custom.

Only acts when size is Custom.

Moves: hook.

custom_t_hook_gap_side_rounding

Hook slot corner rounding



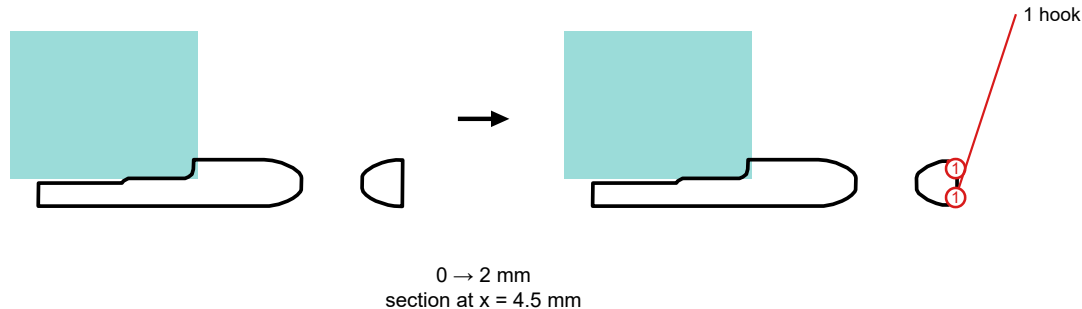
This dial changes no shape: the J-hook cord catch is drawn without this dial; it only appears in the Cutouts Only 2D debug overlay.

Changes no shape of the printed tool, because the J-hook cord catch draws its own corners.

Default 0 · Range 0 to 3 · Step size 0.1 · Unit mm

custom_t_hook_top_bottom_rounding

Hook edge rounding



Two vertical slices of the one-sided puller at x = 4.5 mm, before left and after right, the top face up. Left: the defaults with hand size set to Custom, the plug in teal in the cut. Right: hook edge rounding at 2 mm: flares the top and bottom edges of the cord hook's slot and crossbar. Marked in red: 1, the hook: the cord hook slot in the bottom edge. The body edge, the pocket, the seat and the wall notch stay where they were.

Flares the top and bottom edges of the cord hook's slot and crossbar.

Default 0 · Range 0 to 3 · Step size 0.1 · Unit mm

Before: 0. After: 2. Context: size = Custom. Section at x = 4.5 mm.

Only acts when size is Custom.

Moves: hook.

Measuring guide

Everything here is in mm: a US plug is about 25 mm wide, so a 1 on your paper means you measured in inches. You need a caliper or a ruler with mm marks, the plug in its outlet, and your own hand only if you pick Measure my hand. Print the measuring form at 100 % and fill it in as you go: the sections below are the form's rows, in the same order, and the card names (R1, C1, F1 / F2) are the cards of the printed measuring stencil.

2. Plug length

Customizer name `measure_plug_length`, row 2 of the form.

With the plug in the outlet, ruler from the wall plate face to the plug's back face. On a laptop or charger plug, measure from the device's edge.

Typical: 20 to 50 mm. Example: 38 mm.

With the stencil: card R1.

3. Plug width at the prong end

Customizer name `measure_plug_width_prong_end`, row 3 of the form.

Caliper straight across the plastic body just behind the prongs, the wide way, parallel to the wall. Measure the body, not the metal prongs; if the head flares, take the widest part of that first stretch.

Typical: 20 to 38 mm. Example: 34 mm.

With the stencil: card R1.

4. Plug width at the cord end

Customizer name `measure_plug_width_cord_end`, row 4 of the form.

The same wide direction, but where the cord leaves the plug body. Skip any soft rubber strain relief and measure the last part of the hard body you would grip.

Typical: 10 to 35 mm. Example: 13 mm.

With the stencil: card R1.

5. Plug thickness at the prong end

Customizer name `measure_plug_thickness_prong_end`, row 5 of the form.

Caliper across the plug body's thin direction, usually top to bottom on a flat plug, just behind the prongs. A plug 24 mm thick or more needs the two-sided puller, so measure carefully.

Typical: 12 to 30 mm. Example: 16 mm.

With the stencil: card R1.

6. Plug thickness at the cord end

Customizer name `measure_plug_thickness_cord_end`, row 6 of the form.

The same thin direction, where the cord leaves the plug body, skipping the soft strain relief. On a round extension-cord plug this can be as fat as the prong end.

Typical: 8 to 30 mm. Example: 9 mm.

With the stencil: card R1.

7. Cord thickness

Customizer name `measure_cord_thickness`, row 7 of the form.

Caliper across the cord just behind the plug, on its thin side: flat cord the narrow way, round cord the diameter. With the stencil, the smallest C1 slot that slips over the cord is this number.

Typical: 3 to 7 mm. Example: 5 mm.

With the stencil: card C1.

8. Outlet cover plate style

Customizer name `measure_wall_plate_style`, row 8 of the form.

Look at the outlet cover plate: two small oval openings is Standard flat plate; one big rectangle per outlet is Rocker / Decora; a bigger, thicker plate is Oversized / Jumbo; no plate or a flush outlet is No plate / flush.

The choices: Standard flat plate, Rocker / Decora, Oversized / Jumbo, No plate / flush.

11. Finger width

Customizer name `measure_finger_width`, row 11 of the form.

Caliper across the widest knuckle of your middle finger, the finger that goes into the pull hole. With the stencil, find the smallest F1 or F2 hole your finger passes through comfortably and subtract 5 from its number.

Typical: 16 to 24 mm. Example: 22 mm.

With the stencil: card F1 / F2.

No caliper? Take a ring that fits that finger snugly, measure the ring's inner diameter in mm and add 1.5 mm.

12. Hand width

Customizer name `measure_hand_width`, row 12 of the form.

Caliper or ruler straight across the four knuckles of your flat hand, fingers together, no thumb. Measure at the widest point.

Typical: 70 to 100 mm. Example: 88 mm.

With the stencil: card R1.

15. Strap width

Customizer name `strap_width`, row 15 of the form.

Read the width on the strap's packaging. ONE-WRAP comes in 10, 13, 16, 20 and 25 mm.

Typical: 10 to 25 mm. Example: 15 mm.

Sanity check: each plug width is a two-digit number, roughly 12 to 45, and the prong-end width is usually the bigger one; the finger width is roughly 14 to 32. A number like 1.3 is inches: measure again with the mm side.

When you measure for someone else, a relative or a client, measure their hand for the finger and hand rows and their outlet and plug for the plug rows. Doubtful between two values? Round up: a slightly roomy fit works, a tight one does not.

Measuring form

The next page is the measuring form: print it at 100 % (actual size, never fit to page) and check the 50 mm bar with a ruler before you trust it. Fill in the blanks top to bottom, then type the numbers into the Customizer in the same order. Its text twin is `measuring-form.md` beside this packet's twins.

Plug Puller — Measuring form, one-sided puller

Measure in mm. Fill in the blanks top to bottom, then type them into the Customizer in the same order.

Print at 100 %

50 mm

- ① **plug_preset**
Plug preset
☒ **Measure my plug** ☐ Flat 2-prong lamp plug - NEMA 1-15
☐ Standard 3-prong plug - NEMA 5-15
☐ Heavy-duty extension cord - NEMA 5-15
☐ Wide 2-prong appliance plug - NEMA 1-15
- ② **measure_plug_length**
Plug length
default 25.5 mm → _____ mm card R1
- ③ **measure_plug_width_prong_end**
Plug width at the prong end
default 25 mm → _____ mm card R1
- ④ **measure_plug_width_cord_end**
Plug width at the cord end
default 25 mm → _____ mm card R1
- ⑤ **measure_plug_thickness_prong_end**
Plug thickness at the prong end
default 20 mm → _____ mm card R1
- ⑥ **measure_plug_thickness_cord_end**
Plug thickness at the cord end
default 20 mm → _____ mm card R1
- ⑦ **measure_cord_thickness**
Cord thickness
default 4 mm → _____ mm card C1
- ⑧ **measure_wall_plate_style**
Outlet cover plate style
☒ **Standard flat plate** ☐ Rocker / Decora
☐ Oversized / Jumbo ☐ No plate / flush
- ⑨ **show_plug_preview**
Show the see-through plug
☒ **leave on**
- ⑩ **size**
Hand size
☐ Small ☒ **Medium** ☐ Large ☐ Measure my hand
☐ Custom
- ⑪ **measure_finger_width**
Finger width
default 20 mm → _____ mm card F1 / F2
- ⑫ **measure_hand_width**
Hand width
default 85 mm → _____ mm card R1
- ⑬ **attachment**
Attachment
☐ Zip ties ☐ Velcro strap ☒ **Zip ties + Velcro**
☐ None
- ⑭ **velcro_style**
Strap opening style
☒ **Wing** ☐ Classic slot
- ⑮ **strap_width**
Strap width
default 15 mm → _____ mm
- ⑯ **hook_hand**
Hook side
☒ **Right** ☐ Left

